ATLAS Paper Draft
TRIG-2021-01
Version 2
Target journal: JINST

Comments are due by: 21 November 2024

Supporting internal notes

Support Note: https://cds.cern.ch/record/2777971/

Implementation and ~~Performance~~ performance of the Neural-Ringer ~~Algorithm Selecting Electrons at the High Level Trigger~~ electron selection algorithm in the high-level trigger of the ATLAS ~~Experiment~~experiment

Electron trigger identification is a crucial input to many ATLAS physics analysis. To cope with the increase ~~of~~ in peak LHC luminosity from 2017 to 2018~~that pushed~~ , which raised the number of ~~proton-proton~~ proton–proton interactions per beam-crossing to about 60, trigger algorithms were optimized to keep the background trigger rates low while retaining ~~a~~ high efficiency for physics analyses. In the second half of 2017, the Neural-Ringer algorithm, ~~which is~~ based on a set of ~~Multi-Layer Perception (MLP)~~ multi-layer perceptron models, replaced the cut-based strategy used in the FastCalo decision step in the high-level trigger and became the baseline algorithm for triggering events containing electrons with transverse energy above 15~~GeV in the High Level Trigger (HLT),~~ GeV. This was done as part of the ATLAS online system ~~CPU~~ CPU-time reduction campaign. It was the first time a ~~neural network~~ neural-network method was employed as a baseline ~~selection~~ algorithm for event selection in the ATLAS trigger system~~which results in a higher background early rejection strategy,~~ , and resulted in the early rejection of at least 13 times ~~with respect to~~ more background than the cut-based ~~, while maintaining electron~~ strategy, while leaving the electron selection efficiency nearly unchanged.

Document created on 24th November 2024 using ATLAS LaTeX Version 15.4.0.

Analysis Team

[*email:* atlas-TRIG-2021-01-analysis-team@cern.ch]

Jose Seixas, Denis Oliveira Damazio, Chris Meyer, Werner Spolidoro Freund, Joao Victor Da Fonseca Pinto, Micael Verissimo De Araujo

ATLAS Paper

TRIG-2021-01

24th November 2024

Journal: JINST

Draft version 2

Implementation and ~~Performance~~ performance of the Neural-Ringer ~~Algorithm Selecting Electrons at the High Level Trigger~~ electron selection algorithm in the high-level trigger of the ATLAS ~~Experiment~~experiment

The ATLAS Collaboration

Electron trigger identification is a crucial input to many ATLAS physics analysis. To cope with the increase ~~of~~ in peak LHC luminosity from 2017 to 2018~~that pushed~~ , which raised the number of ~~proton-proton~~ proton–proton interactions per beam-crossing to about 60, trigger algorithms were optimized to keep the background trigger rates low while retaining ~~a~~ high efficiency for physics analyses. In the second half of 2017, the Neural-Ringer algorithm, ~~which is~~ based on a set of ~~Multi-Layer Perception (MLP)~~ multi-layer perceptron models, replaced the cut-based strategy used in the FastCalo decision step in the high-level trigger and became the baseline algorithm for triggering events containing electrons with transverse energy above 15~~GeV in the High Level Trigger (HLT),~~ GeV. This was done as part of the ATLAS online system ~~CPU~~ CPU-time reduction campaign. It was the first time a ~~neural network~~ neural-network method was employed as a baseline ~~selection~~ algorithm for event selection in the ATLAS trigger system~~which results in a higher background early rejection strategy,~~, and resulted in the early rejection of at least 13 times ~~with respect to~~ more background than the cut-based ~~, while maintaining electron~~ strategy, while leaving the electron selection efficiency nearly unchanged.

Contents

1 Introduction

The ATLAS physics ~~program~~ programme relies on an efficient trigger system [1] to record a highly ~~signal-rich subset of all~~ signal-enriched subset of collision events produced by the Large Hadron Collider (LHC) at CERN. Electron triggers [2] are important for searches for new particles, ~~the measurement~~ measurements of Standard Model cross-sections, and ~~in the precise measurement~~ precise measurements of the properties of fundamental particles such as the Higgs [3, 4] and ~~W~~ W bosons.

A hierarchical two-level trigger system is used to select events ~~with electrons produced by the LHC~~ of interest produced in LHC beam collisions. The first-level (L1) trigger, implemented with custom-made electronics located on the detector, ~~utilises coarser granularity from the calorimeters~~ utilizes the calorimeters at coarser granularity to reduce the event rate from the 40~~MHz bunch crossing~~ MHz bunch-crossing rate to below 100 kHz, of which about 35% ~~correspond~~ corresponds to events with high-energy electromagnetic showers, potentially coming from electrons. The L1 ~~level~~ trigger also defines regions-of-interest (RoIs) [5], situated around local regions with high transverse energy.

The RoIs ~~accepted by~~ defined by the L1 trigger are processed by the ~~High Level Trigger~~ high-level trigger (HLT) implemented in software. The HLT operates ~~from~~ in a large computing farm and should reduce the ~~number~~ rate of events written to disk to an average ~~rate~~ of about 1 kHz. For electron selection, the HLT is split into two stages: a fast but efficient computation followed by ~~a~~ precise processing, ~~which employs offline inspired algorithms that are~~ where the latter employs offline-inspired algorithms adapted to the harsh conditions (e.g.~~,~~ time and memory constraints) of the trigger system.

The schematic diagram ~~of~~ in Figure 1 ~~represents~~ shows the chain of ~~algorithm~~ algorithms for electron reconstruction in the HLT~~of electron reconstruction. The HLT~~. Electron selection starts with ~~the fast calorimeter reconstruction~~ a fast calorimeter-reconstruction step (FastCalo), which has two implementations: ~~an~~ the original cut-based selection and ~~a new apporch based on~~ the new neural-network (NN) approach using the Neural-Ringer algorithm presented in this paper. This ~~processing step~~ is followed by ~~the fast track reconstruction~~ a fast track-reconstruction step (FastElectron), which ~~applies restrictions~~ also places requirements on variables that indicate the quality of proximity matching between the electron candidate positions reconstructed from the track and the ~~calorimeter. The selection of particle candidates by the HLT is performed~~ calorimetric measurements. The HLT tries to select particle candidates at each step~~, so that~~; if it fails at a certain step, subsequent steps are not executed. This is essential in order to reduce the CPU resources and time needed by the HLT to reconstruct the event and make a trigger decision.

This paper describes the performance improvements due to the introduction of the Neural-Ringer algorithm into the HLT fast ~~calorimenter reconstruction step of the electron triggers, which~~ calorimeter-reconstruction step for electron triggers. It became the baseline ~~selection of~~ algorithm for selecting events containing at least one isolated electron with transverse energy above 15 GeV in the 2017 ~~proton-proton~~ proton–proton collision ATLAS data-taking ~~. The~~ at $\sqrt{s} = 13$ TeV in LHC Run 2. Full details of its ~~identification procedure are in Section 2~~electron identification procedure, including the training and tuning strategies~~.~~ ~~The online~~ , are given in Section 2. Online performance results from the Neural-Ringer and ~~the~~ cut-based strategies are presented in Section 3. ~~A comparison of the~~ The Neural-Ringer ~~with the~~ and cut-based ~~triggers~~ electron triggers are compared from an offline perspective using statistical methods ~~through the offline perspective is performed~~ in Section 4. Finally, the conclusions and prospects are ~~derived~~ presented in Section 5.

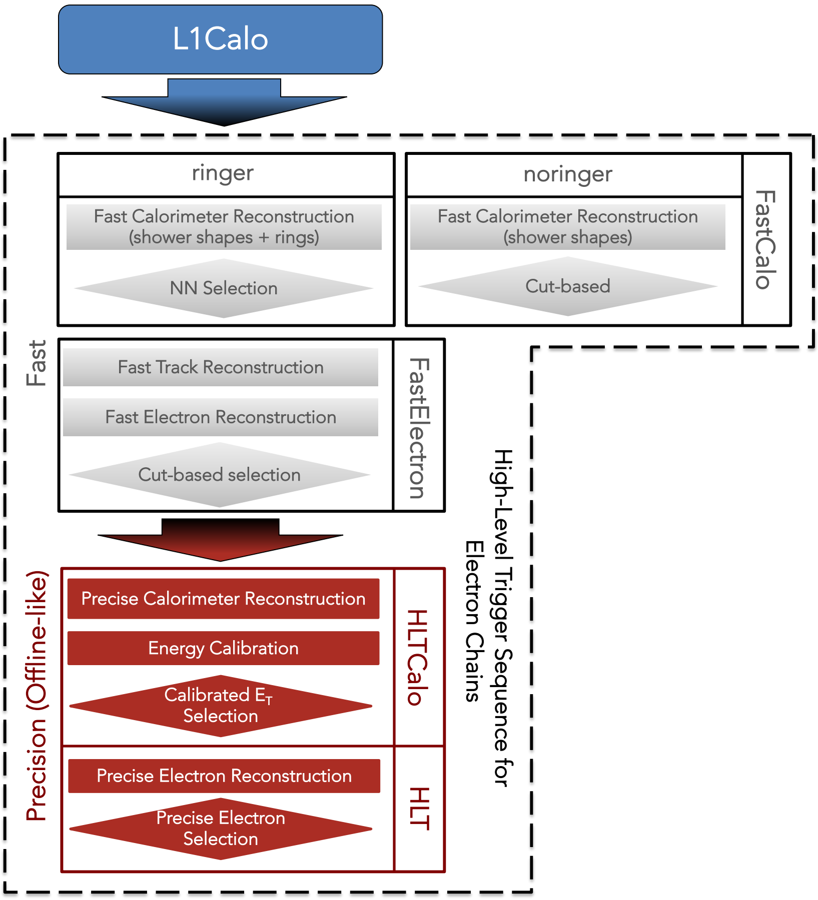


Figure 1: Chain of algorithms for electron triggers in the HLT. ~~Difference~~ Differences between the original baseline (no-ringer) and new (ringer) electron triggers lie ~~on~~ in the first step, ~~the~~ called FastCalo. ~~Gray~~ Grey blocks represent algorithms in the fast stage, and red blocks represent ~~precision step~~ precision-step algorithms.

2 The Neural-Ringer ~~Algorithm~~algorithm

Instead of using electron ~~shower shapes~~ shower-shape variables [1] to discriminate ~~electrons from~~ between electrons and background, the Neural-Ringer algorithm ~~adapt another shower representation made of transverse energy rings built from the sum of calorimeter cells~~ adopts a shower representation consisting of rings of transverse energy, built from sums of calorimeter-cell energies, around each candidate provided as an RoI by the L1 trigger. The distribution of those energy sums is used by ~~an~~ a set of neural networks to ~~classify~~ distinguish electrons from background. This section ~~is dedicated to~~ describes the Neural-Ringer algorithm.

2.1 The ATLAS ~~Calorimeter~~calorimeters

The ATLAS ~~uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z-axis along the beam-pipe. The x-axis points from the IP to the centre of the LHC ring, and the y-axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse plane,~~

~~ϕ being the azimuthal angle around the beam-pipe. The pseudorapidity is defined in terms of the polar angle θ as η = -ln $tan(\theta/2)$. The angular distance ΔR is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Transverse momenta and energies are defined as $pT = p \sin\theta$ and $E_T = E \sin\theta$, respectively.~~

~~The ATLAS~~ calorimeters comprise rectangular cells in the $\eta \times \phi$ plane[1] for different layers in depth, except for the forward calorimeters. The calorimeter system covers the pseudorapidity range $|\eta| < 4.9$ [6]. Within the region $|\eta| < 3.2$, ~~an electromagnetic calorimetry (ECAL)~~ electromagnetic calorimetry is provided by ~~the~~ a high-granularity lead/~~Liquid-Argon~~ liquid-argon (LAr) ~~calorimeters constituted~~ calorimeter (ECal) composed of barrel (EMB1,2,3) ~~and end-cap~~ sections covering $|\eta| < 1.475$ and endcap (EMEC1,2,3) sections covering $1.375 < |\eta| < 3.2$, with an additional thin LAr presampler (PS) covering the barrel (PreSamplerB) and ~~end-cap~~ endcap (PreSamplerE) ~~, in order~~ regions to correct for energy losses in ~~material upstream of the calorimeters~~upstream material [7, 8]. ~~The Hadronic calorimetry (HCAL)~~ Hadronic calorimetry within $|\eta| < 3.2$ is provided by ~~the~~ a steel/~~scintillating-tile~~ scintillator-tile calorimeter (TileCal) [9, 10] ~~, constituted~~ composed of three barrel ~~structures~~ sections (TileBar0,~~TileBar1 and TileBar2) within $|\eta| < 1.0$~~1,2) covering $|\eta| < 1.0$ and three extended barrel ~~structures~~ sections (TileExt1,2,3) covering ~~the region of~~ $0.8 < |\eta| < 1.7$, and two copper/LAr hadronic endcap calorimeters (HEC) [11] ~~within $1.5 < |\eta| < 3.2$ divided in~~ divided into four modules (HEC0,1,2,3) covering $1.5 < |\eta| < 3.2$. Transition regions between calorimeters are used ~~to locate~~ as locations for detector services and induce ~~a sharing of showers between calorimetersthat~~ shower-sharing between calorimeters, which degrades energy measurements in those regions. Here, the ~~transition from the barrel to the endcap~~ barrel-to-endcap transition is complemented by the intermediate tile calorimeter (ITC)~~composed by~~, which is composed of two modules (TileGap1,2,3) that are ~~both attached on the face of the external barrel~~ attached to the external faces of the barrel and are used to estimate signal losses [11]. The ~~solid angle~~ solid-angle coverage is completed with forward copper/LAr and tungsten/LAr calorimeter (FCal) modules ~~optimised~~ optimized for electromagnetic (EM) and hadronic (HAD) measurements, respectively. The ~~specified calorimeters provide~~ calorimeter system provides full azimuthal (ϕ) coverage with a total of about 190~~,~~000 readout cells.

The electromagnetic and hadronic calorimeters are segmented [6] [2] longitudinally (i.e. in depth) into three~~sampling layers~~ [2] sampling layers, where each layer has its own lateral ($\eta \times \phi$) granularity. ~~Besides granularity, the amount of detector~~ The amount of material in front of ~~the detector in~~ each sampling layer ~~also~~ varies with $|\eta|$, leading to variations in the expected lateral and longitudinal ~~profiles. Additionally, the expected profile is also dependent~~ shower profiles. The expected profiles also depend on the physics object's total energy. In the EM calorimeter, most of the EM shower energy is collected in the second layer (EM2), while the third layer (EM3) provides measurements of energy deposited ~~in~~ by the shower tails. ~~Although, it is important to mention~~ However, it should be noted that the first EM layer (EM1) plays an important role in ~~the discrimination of electrons against~~ discriminating between electrons and π^0 mesons. The hadronic calorimeters, which surround the EM detectors, provide additional electron discrimination power through ~~further~~ energy measurements of possible EM shower tails, as well as rejection of events with activity of hadronic origin ~~with~~ via the three sampling layers (HAD1,2,3).

[1] ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z-axis along the beam-pipe. The x-axis points from the IP to the centre of the LHC ring, and the y-axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the beam-pipe. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln\tan(\theta/2)$. The angular distance ΔR is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Transverse momenta and energies are defined as $p_T = p\sin\theta$ and $E_T = E\sin\theta$, respectively.

[2] ~~The two central layers in HCAL end-cap are summed to result in a single measurement.~~

[2] The two central layers of each hadronic endcap calorimeter are summed to provide a single measurement.

2.2 Building the ~~Rings~~rings

~~The~~ Since the Neural-Ringer algorithm was ~~considered~~ designed to operate online ~~at~~ in software at the FastCalo stage~~in software~~, approximations were made. Specifically, ~~the~~ rectangular rings are constructed to match the grid layout of the calorimeter cells, and the ring coverage is bounded by the RoI's area. In addition, granularity changes ~~from the cell sizes from~~ due to different cell sizes in different regions of the detector are neglected in order to employ a grid-like algorithm. ~~Here,~~ When considering only the EM1 layer, the η~~granularity~~ granularity used by the algorithm is 0.003 for the entire ~~eta~~ η range defined ($|\eta| < 2.5$)~~.~~ ~~However, the detector construction at this layer cover the region of~~, although this layer actually covers the region $0 < |\eta| < 2.37$ and has ~~different granularities in~~ three different η~~: starts with~~ granularities: 0.003 ~~at~~ for $0 < |\eta| < 1.81$~~;~~, 0.004 ~~at~~ for $1.81 < |\eta| < 2.01$ and 0.006 at $2.01 < |\eta| < 2.37$.

~~Here, the algorithm~~The algorithm's RoI covers a ~~region of the calorimeter, centered in the L1 provided RoI, of~~ calorimeter region of 0.4×0.4 in the $\eta \times \phi$ plane, centred on the L1-provided RoI. The algorithm starts ~~on~~ at the second EM sampling layer (EM2), where ~~position given in~~ the $\eta \times \phi$ ~~plane~~ position of the cell with ~~highest E_T on the algorithm~~the highest E_{T}in the algorithm's RoI is taken as the ~~center~~ centre of the cascade interaction in the calorimeter. The initial ~~seed cell~~ seed-cell position is propagated to other calorimeter layers in order to define the axis of the rings in ~~that layer~~those layers. Then, for each layer, the algorithm refines the seed position ~~searching for~~ by searching for the most energetic cell inside ~~of the~~ a reconstruction window centred on the EM2 initial seed, and the energy deposition of an incoming particle is extracted by building concentric rings of cells, ~~or simply "rings"~~simply referred to as 'rings'.

The refined seed (~~$c_{hot,l}$~~$c_{\mathrm{hot},l}$) is used as the starting point to build the rings. A ring $R_{n,l}$ contains all $C_{n,l}$ cells in calorimeter layer l which are n cells away from the refined seed (~~See~~ see Figure 2 for an illustration of the parameters). Formally,

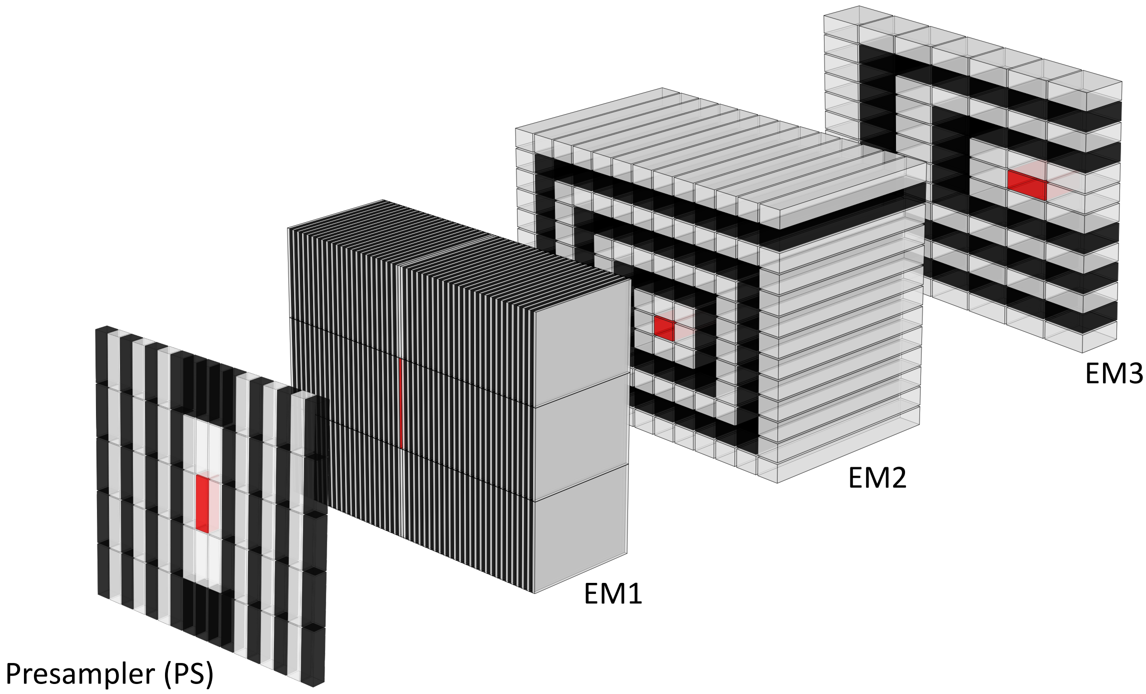
$$\cancel{R_{n,l} = \left\{ c_{n,l} \mid n = \left\lfloor \max\left(\frac{|\eta_{i,l} - \eta_{hot,l}|}{h_{\eta,l}}, \frac{|\phi_{i,l} - \phi_{hot,l}|}{h_{\phi,l}} \right) \right\rceil, \forall c_{i,l} \in \Theta_{RoI,l} \right\},}$$

$$R_{n,l} = \left\{ c_{n,l} \mid n = \left\lfloor \max\left(\frac{|\eta_{i,l} - \eta_{\mathrm{hot},l}|}{h_{\eta,l}}, \frac{|\phi_{i,l} - \phi_{\mathrm{hot},l}|}{h_{\phi,l}} \right) \right\rceil, \forall c_{i,l} \in \Theta_{\mathrm{RoI},l} \right\},$$

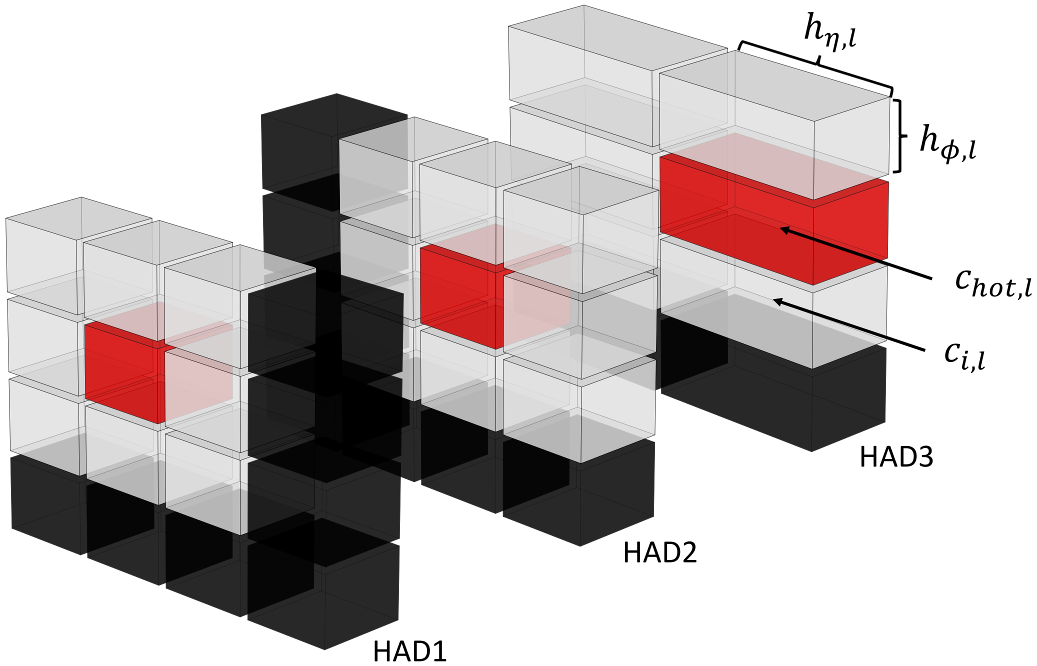
where (~~analogous to~~ and analogously for ϕ when ~~suitable):~~ appropriate) $\eta_{i,l}$ and ~~$\eta_{hot,l}$~~ $\eta_{\mathrm{hot},l}$ are respectively the $c_{i,l}$ and ~~$c_{hot,l}$ cellsposition~~ $c_{\mathrm{hot},l}$ cells' positions in η;$h_{\eta,l}$ is the ~~l^{th} layer~~l^{th} layer's cell size in η; ~~$\Theta_{RoI,l}$~~ $\Theta_{\mathrm{RoI},l}$ is the set of cells in the ~~l^{th}~~ l^{th} layer which are within the Ringer RoI; and $\lfloor \cdot \rceil$ is the *round* function. Table 1 shows the number of rings computed at each layer and the respective granularity.

The ~~sum~~ sums of the transverse energies of cells $c_{n,l}$ in the ~~ring~~ rings $R_{n,l}$ form a vector of discriminating quantities. They are ordered ~~outwards and~~ laterally outwards and longitudinally from the innermost to outermost layer of the calorimeter~~and provide~~, thus providing the ring-shaped information ~~of~~ about the RoI.

The ~~ring building~~ ring-building process results in a total of 100 rings ~~in total~~ across all calorimeter layers in each RoI. ~~Thereby, a dimensionality reduction is provided~~ Well-known features of EM shower physics allow a large reduction in dimensionality to be achieved by compacting the typical input space ~~dimensionality~~ of approximately 1000–1200 cells per ROI into ~~the above-mentioned~~ these 100 rings~~through the usage of EM shower physics knowledge~~: the rings can keep a complete description of the ~~symmetric lateral and longitudinal~~ longitudinal and symmetric lateral information. However, the algorithm only approximates



(a) ~~Eletromagnetic~~ Electromagnetic calorimeter cells within the ringer reconstruction window.



(b) Hadronic calorimeter cells within the ringer reconstruction window.

Figure 2: Sketch ~~to illustrate~~ illustrating the ring-shaped energy description. The most energetic cell is in red, while the consecutive neighbouring rings are represented by alternating ~~gray~~ grey and black cells.

Table 1: Nomenclature defining the Neural-Ringer algorithm layers and sections, as well as the respective parameters employed and the calorimeter sampling layers from which the cells are extracted. The Neural-Ringer algorithm parameters were dependent on the ringer layer l ~~and~~ but independent ~~on~~ of η and E_{T} during ~~Run 2.~~ LHC Run 2. The parameters are the ring size in η ($h_{\eta,l}$), ϕ ($h_{\phi,l}$) and the number of rings to be computed in each layer (~~N_l~~ N_l).

Ringer		Calorimeter Sampling			Parameters		
Section	Layer (l)	Barrel	ITC	End-cap	$h_{\eta,l}$	$h_{\phi,l}$	N_l
EM	PS	PreSamplerB		PreSamplerE	0.025	0.1	8
	EM1	EMB1		EMEC1	0.003	0.1	64
	EM2	EMB2		EMEC2	0.025	0.025	8
	EM3	EMB3		EMEC3	0.050	0.025	8
HAD	HAD1	TileBar0 TileExt0	TileGap3	HEC0	0.1	0.1	4
	HAD2	TileBar1 TileExt1	TileGap1	HEC1 HEC2	0.1	0.1	4
	HAD3	TileBar2 TileExt2	TileGap2	HEC3	0.2	0.1	4

this concept in order to meet the online operation requirements and avoid further manipulation of the ~~instrumented~~ instrumental information. Figure 3 shows the ring-shaped mean profile differences between electrons and jet particles.

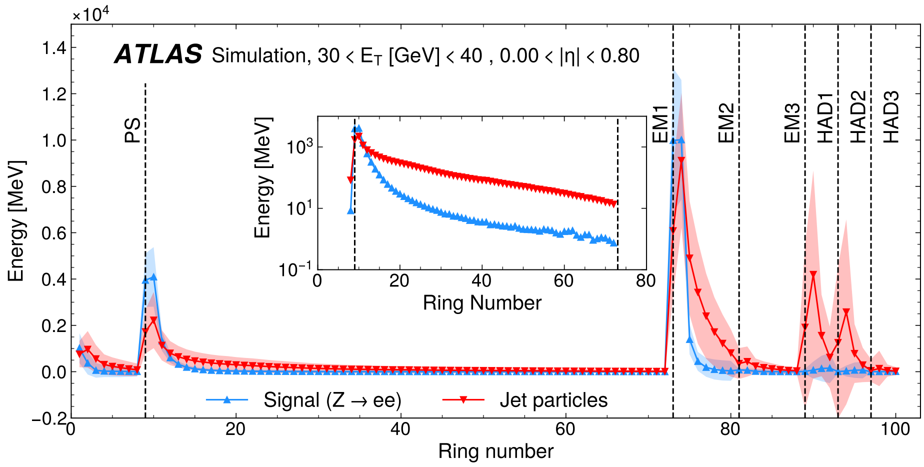


Figure 3: Average ring profile from ~~collision~~ simulation data at ~~$30 \le E_T < 40$~~ $30 \le E_{\mathrm{T}} < 40$ GeV and $0.0 \le \eta < 0.8$ for each layer of the calorimeter. The ~~embedded~~ profile in the inset shows the ~~EM1 layer~~ EM1-layer ring energies ~~in~~ on a logarithmic scale. Most ~~part~~ of the energy ~~for~~ of the electrons is absorbed by the ~~eletromagnetic~~ electromagnetic layers. For the jets, most of the energy is absorbed by the hadronic layers.

2.2.1 Ring ~~Sums~~ sums as ~~Discriminant Variables~~discriminating variables

The current strategy concatenates all rings ~~in~~ into a single vector of 100 variables. ~~An absolute normalization using~~ The variables are normalized to the absolute value of the total RoI energy~~is applied to normalize the variables. This procedure was initially proposed and examined by [12], as a way of preserving the shower~~

~~energy profile in fractions of the total energy.The absolute value is used to avoid reflecting the values along the axis due to negative noise accumulation, a behavior which would impact the physics representation of the normalized values and require a more complex decision boundary~~, as shown in Eq. (1):

$$r'_k = \frac{r_k}{|\sum_{i=1}^{100} r_i|}, \quad \forall\, k \in \{1, \ldots, 100\}., \tag{1}$$

~~Where~~ where r_k or r_i is the sum of transverse energy of the cells in the ring with index k or i, and r'_k is the normalized ring value with index k. ~~It is specially valuable for its simplicityas it uses of a non parametric~~ The absolute value is used because of negative noise accumulation from some cells within the reconstruction window that were not excited by the particle shower. This procedure was initially proposed and examined in Ref. [12] as a way of preserving the shower energy profile as fractions of the total energy. It is especially valuable because of its simplicity, since it uses a non-parametric approach, and for allowing easy interpretation of the shower profile.

2.3 Motivation for using a ~~Set~~ set of ~~Neural Networks~~neural networks

The normalized concatenated ring vector ~~feeds~~ is fed into a set of ~~Multi Layer Perceptrons (MLP). A single model is drawn from the set for operation based on~~ multi-layer perceptrons (MLPs). Each MLP corresponds to a model specifically trained for a particular region of $E_{\mathrm{T}} \times |\eta|$space. The model chosen for each shower's vector is the one trained for the nearest region ~~on the spacefor which it was specific trained~~of phase space.

~~In regards to~~ For calorimetry, the ~~usage~~ use of a set ~~with specific modelsper phase space region allows to mitigate~~ of models, each specific to a particular phase-space region, mitigates fluctuations in the ~~variable~~ variables' profiles due to ~~the detector responseand energy differences of the showers~~differences in shower energy and detector response. A large contribution comes from ~~granularity changes, which are discrete~~ discrete detector-granularity changes in the phase space. Other important factors influencing such profiles are the amount of the detector material as a function of $|\eta|$ and the ~~dependency of underlying processes in the shower development with respect to the incoming particle~~dependence of the underlying shower-development processes on the incoming particle's energy. Although ~~these alterations~~ the latter changes are mostly continuous, it is also possible to approach the problem ~~using~~ via space discretization.

At the hardware resource level, splitting the training into multiple small models to accommodate ~~inhomogeneity of the detector response~~ detector response inhomogeneities speeds up the training cycle and allows ~~to handle data~~ data to be handled in memory, since only a small portion of the data is used to train ~~that~~ a specific model. Limiting the memory ~~requirement~~ required for tuning the models ~~is particularly interesting~~ was of particular interest in order to benefit from low-memory processing nodes, ~~which were predominantly available in WLCG~~as these were the ones mostly available in the Worldwide LHC Computing Grid [13] during our ~~developments~~development effort. In addition, the decomposition of the problem can also lead to a reduction ~~of the~~ in training time [14], as observed heuristically when comparing the tuning time of the set ~~to a single model. Regarding the~~ with that of a single overall model. With regard to operation, the proposed set only requires the choice of a single model ~~to operate instead of a more often committee machine approaches that~~ in order to operate, whereas the more common committee-machine approaches require the computation of different models to run in parallel with a fusion method [15]. Therefore, the chosen set configuration adds minimal overhead in terms of trigger latency.

Similarly, ~~the~~ our choice was to ~~operate with~~ use low-complexity models, with a single hidden layer ~~with~~ and few neurons, to ~~obtain a competitive method with respect to~~ develop a method competitive with the cut-based approach ~~,~~ in terms of processing speed.

2.4 Training and ~~Decision Making~~ decision making for 2017 and 2018 ~~Operation~~operation

The training procedure for 2017 operation was ~~built with~~ based on samples of simulated $Z \to ee$ ~~decay, selected by the~~ decays, selected with a $Z \to ee$ ~~method~~tag-and-probemethod [1], and background. For simulated background samples, a filter was applied to enrich them with particles produced in the hard scatter with a summed transverse energy exceeding 17 GeV in a region of $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$. During 2016 ~~data taking, the~~ data-taking, pile-up reached ~~was~~ 40 interactions per bunch crossing. ~~Here, both simulated samples include a pile-up~~ Both types of simulated samples included an average of 60 ~~interactions in average,~~ pile-up interactions per bunch crossing, which ~~had never been seen before~~ was not seen in collision data until the end of 2017, when the ~~observed~~ peak pile-up exceeded 60. Recently, in 2023, the highest observed level of pile-up ~~observed~~ exceeded 60 again.

After the training stage, ~~thresholds on the discriminating score~~ threshold values for the discriminating variables must be determined ~~for selection of~~ in order to select good electron candidates. To set the correct ~~value~~ values without causing any ~~inefficiency in the end of the HLT in~~ HLT inefficiency for collision data, probe electrons ~~, selected by~~ selected by the $Z \to ee$ ~~method and approved by the tightest offline available criteria, from full~~ tag-and-probemethod from the full 13 TeV pp ~~data~~ collision dataset recorded by ATLAS in 2016 ~~with LHC operating at a centre-of-mass energy of $\sqrt{s} = 13$ TeV were used~~ were used if they also satisfied the tightest available offline criteria. For background, objects rejected by the $Z \to ee$ ~~method, that~~ tag-and-probemethod, which do not belong to any ~~tag-probe~~ tag–probe electron pair, ~~and reproved~~ were used if they were also rejected by the loosest ~~offline available criteria were used. For the~~ available offline criteria. For 2018 operation, a purely data-driven strategy was used, selecting events from the 2017 recorded collision data to train the models and adjust all thresholds. ~~The~~ To be used for tuning and analysis~~performed on collision data recorded have their quality ensured by the~~, the recorded collision data were first required to pass the standard ATLAS data quality procedure.

2.4.1 Training and ~~Decision Making~~decision making

The 2017 procedure derived ~~specific MLP models~~ a specific MLP model for each region ~~on~~ of $E_{\mathrm{T}} \times |\eta|$ ~~space, with~~ phase space, using simulated samples, and defined ~~the discriminant cut~~ threshold values for the discriminating variables by using 2016 collision data and targeting a signal efficiency similar to that of the cut-based trigger. The training procedure and ~~decision making~~ decision-making processes were the same for all ~~phase space~~ phase-space regions. For the Neural-Ringer, each MLP is an expert model for a single ~~phase space~~ phase-space region, containing its own topology and parameters.

The model parameters ~~have been~~ were optimized using events selected ~~on~~ from simulation datasets. The structure ~~is a fully-connected single~~ consists of a single fully-connected hidden layer, which may contain from 5 to 20 hidden units, an input layer with 100 inputs, one for each ring, and an output layer with a single neuron. The activation ~~functions for both~~ function for both the hidden and output neurons ~~are the~~ is a hyperbolic tangent, which has ~~been often~~ often been used in MLP designs [16].

To assess the statistical fluctuations of the efficiency measurements, the stratified ~~k-fold ($k = 10$)~~ k-fold cross-validation method [16]~~was used at the price of increased computational cost by repeating~~, with

$k = 10$, was used, although repeating the training and testing procedures on different randomly chosen subsets increased the computational cost. The stratified ~~k-fold is among~~ k-fold is one of the most common cross-validation techniques and consists of partitioning the dataset ~~in k~~ into k disjoint subsets, each model being tested with the ~~ith subset and~~ i^{th} subset after being trained with the ~~remaining~~ other ones. The dataset stratification follows the empirical order of the event selection, in order to allow straightforward job parallelization, i.e. random permutation is not employed.

For each cross-validation sort and working point, 100 networks, ~~initiated as from~~initialized as in Ref. [17], are optimized by back-propagating the ~~Mean Squared Error (MSE)~~ mean squared error through the RPROP algorithm [18]. This repeated model initialization ~~aims at avoiding~~ seeks to avoid poor sub-optimal solutions due to the ~~usage~~ use of gradient-based algorithms. For each training procedure, only three models out of the 100 are retained: two for retrieving the best efficiency when fixing the sensitivity or false positive probabilities to the baseline FastCalo efficiency and another ~~resulting in~~ that gives the $\mathrm{SP}_{\mathrm{max}}$ value. The SP value is defined as:

$$\sqrt{\sqrt{P_D(1-F_A)}\cdot\frac{P_D+(1-F_A)}{2}}$$

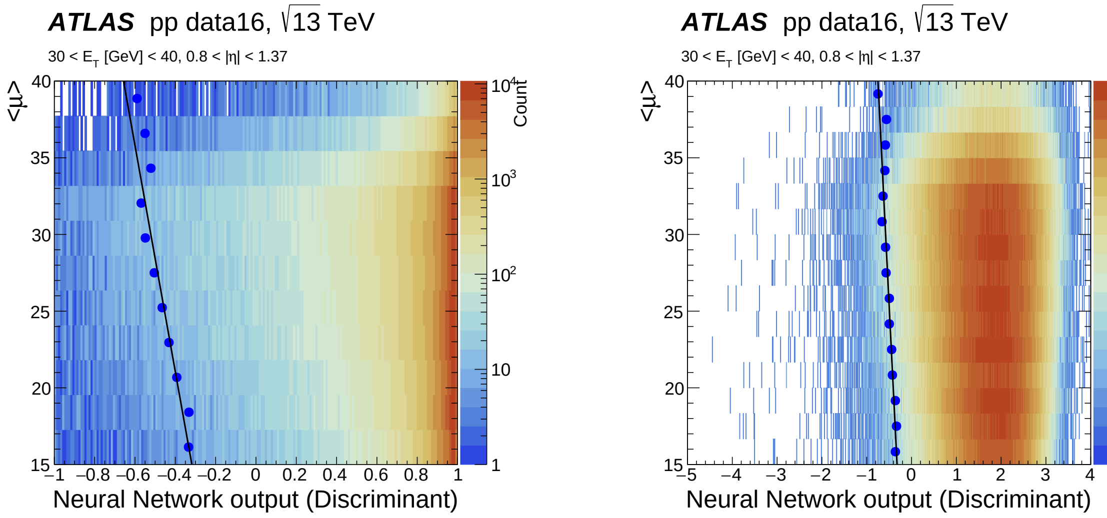
$$\sqrt{\sqrt{P_{\mathrm{D}}(1-F_{\mathrm{A}})}\cdot\frac{P_{\mathrm{D}}+(1-F_{\mathrm{A}})}{2}}$$

where ~~P_D~~ P_{D} is the signal detection probability (or sensitivity) and ~~F_A~~ F_{A} is the probability of a fake signal (false positive). An early stopping [16] technique is employed to avoid model over-training. The selection of the optimal iteration and operating model is based on ~~an~~ a heuristic approach (multi-stop) [19].

Computational resources were used to tune 1.3 M shallow-learning neural networks, which resulted in a set of 20 MLPs (five in E_{T} × four in $|\eta|$) for each one of the four working points used by electron triggers in the HLT. ~~Here~~Thus, the first training strategy considered only four regions in $|\eta|$ ($0.0 \rightarrow 0.8 \rightarrow 1.37 \rightarrow 2.5$) and five ~~region in E_T ($15 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50 \rightarrow \infty$). Except for~~ regions in E_{T}(15 GeV $\rightarrow 20$ GeV $\rightarrow 30$ GeV $\rightarrow 40$ GeV $\rightarrow 50$ GeV $\rightarrow \infty$). Apart from few exceptions, the models in the set employed ~~5~~ five neurons in the hidden layer.

After the training stage, ~~the decision is taken~~ decisions are made by applying a threshold ~~in~~ requirement to the one-dimensional output node of each expert MLP. ~~Inspired by~~ As in the HLT likelihood algorithm, the threshold ~~applied for~~ value used in the Neural-Ringer ~~is also~~ algorithm is computed as a linear function of $\langle\mu\rangle$, the mean number of pile-up interactions per bunch crossing. However, ~~a~~ non-linear ~~behavior~~ behaviour was observed near the asymptote of the output activation function (~~See~~ see Figure 4(a)) ~~, which occurred for the medium and tight operation points~~when using the 'medium' or 'tight' operating point. Heuristically, it was observed that this behaviour can be avoided ~~when~~ by replacing the hyperbolic tangent activation function ~~from~~ in the output neuron ~~by~~ with a linear function (Figure 4(b)) for operation, after the training stage is completed. This strategy ~~provides a linear behavior~~ ensures linear behaviour of the targeted signal efficiency with respect to the online pile-up estimator. To account for Monte Carlo and collision data ~~mis-modeling (See~~ mis-modelling (see Figure 5), the parameters of the linear threshold correction were derived using 2016 collision data and ~~upper-bounded by~~ an upper bound of $\langle\mu\rangle = 40$ ~~collisions (e.g., in case of~~ pile-up ~~higher than upper-bound,~~ interactions (i.e. the corrected threshold value is ~~fixed~~constant for pile-up values above the upper bound).

~~Furthermore, it~~ It was observed that the rings in the region $2.37 < |\eta| < 2.47$ had particular profiles due to the lack of strip cells in this region and ~~demanded~~ required a specific discriminant ~~requirement~~ threshold in order to keep the signal efficiencies close to those of the cut-based triggers. In order to avoid retraining



(a) MLP output with a hyperbolic tangent ~~as~~ activation function in the output neuron ~~w.r.t~~ versus pile-up.

(b) MLP output with a linear ~~as~~ activation function in the output neuron ~~w.r.t~~ versus pile-up.

Figure 4: Neural-Ringer output as a function of pile-up, $\langle\mu\rangle$, for $Z \rightarrow ee$ probes in 2016 collision data selected using the training criteria. The computed threshold per $\langle\mu\rangle$ unit for achieving the ~~target~~ targeted efficiency is shown ~~in blue~~ as a full blue circle. The black line is a linear fit ~~of~~ to the thresholds. The output space is computed using the trained model without modifications in (a), while in (b) the activation function ~~from~~ in the output neuron is replaced ~~by~~ with a linear function.

all models, it was decided to split the last $|\eta|$ region in two, from $1.37 \rightarrow 2.5$ to $1.37 \rightarrow 2.37 \rightarrow 2.5$, and duplicate all models, for this region, to the new bin. Finally, the set used to ~~operated~~ operate in 2017, which ~~is a combination of~~ combines every η boundary with each ~~E_T~~ E_{T} boundary, has a total of 25 MLP models. The set's boundaries are defined ~~by~~ in Table 2.

Table 2: E_{T} and η boundaries used to create the set of neural networks.

Model Adjustment	
E_{T} [GeV] boundaries	$\lvert\eta\rvert$ boundaries
$15 \leq E_{\mathrm{T}} < 20$	$0.00 \leq \lvert\eta\rvert < 0.80$
$20 \leq E_{\mathrm{T}} < 30$	$0.80 \leq \lvert\eta\rvert < 1.37$
$30 \leq E_{\mathrm{T}} < 40$	$1.37 \leq \lvert\eta\rvert < 1.54$
$40 \leq E_{\mathrm{T}} < 50$	$1.54 \leq \lvert\eta\rvert < 2.37$
$E_{\mathrm{T}} \geq 50$	$2.37 \leq \lvert\eta\rvert < 2.47$

~~E_T and η boundaries used to create the set of neural networks.~~

On the other hand, the 2018 training procedure employed a single set structure for all working points and used the 2017 collision data, selected with $Z \rightarrow ee$ tag-and-probe event selection, for training, as was the case for the derivation of the offline and final HLT likelihood models [2]. Most of the procedure was kept unchanged for 2018. The developments for 2018 data benefited from the ~~similarity of data conditions with respect to~~ data conditions being similar to those in 2017. To simplify the training strategy, the selection of three models for each training was abandoned, keeping only the $\mathrm{SP}_{\mathrm{max}}$ model, as the previous approach

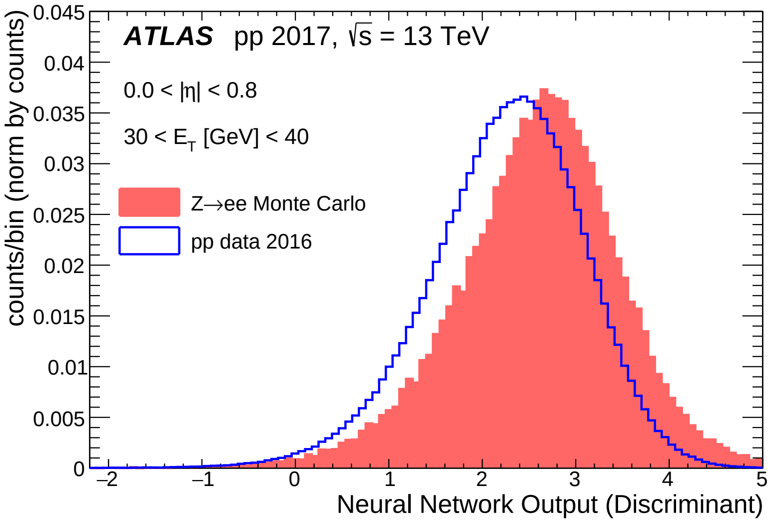


Figure 5: Differences between simulation samples (red) and collision data (blue) obtained in 2016 for ~~a~~ one region of the phase space. The comparison is made between the normalized distributions ~~obtained in both cases for~~ of the output of the neural network (the discriminant) ~~generated~~ obtained for ~~the samples present in the signal set within the selected region of the phase space~~these two cases. The shift ~~on the~~ in discriminant ~~generated by the neural network~~ value between simulation and collision data is caused by ~~the~~ differences ~~of~~ between the ~~simulation-data~~ simulation and data rings used as input ~~into~~ to the model.

was more complex without a clear ~~return on~~ increase in trigger efficiency.

Likewise, MLPs with 5 to 10 units in the hidden layer were tuned, as most models did not require more than 10 units in 2017. The training optimizer was changed to ~~the~~ *Adam* [20], instead of RPROP, and the MLP model topology was modified to ~~:~~ have a hidden layer with ~~a~~ fully connected neurons~~with a~~ , with rectified linear units (RELU) as the activation function, and an output layer with one neuron using ~~the sigmoid as the~~ a sigmoid activation function. With a larger time span for entering operation, specialized MLPs were implemented in the region ~~where a change was detected in the ring profiles~~ $2.37 < |\eta| < 2.47$, where different ring profiles were observed before 2017 operation~~between $2.37 < |\eta| < 2.47$ was added~~, resulting in the operation of 25 MLPs. Finally, the ~~correction limit~~ upper bound for threshold corrections was set to $\langle\mu\rangle = 100$ ~~collisions in other~~ pile-up interactions in order to extrapolate the operation of the models to a ~~never reached~~ never-reached pile-up level~~condition~~.

3 Online ~~Deployment~~ deployment and ~~Performance~~performance

~~During 2017-2018 years (LHC Run 2), the actual~~ The Neural-Ringer ~~implementation has made in four chronological stages~~ algorithm was implemented in four stages during 2017–2018 in LHC Run 2 :

i A development stage up to early 2017, where the Neural-Ringer's potential was estimated from trigger emulation and data reprocessing~~;~~.

ii The commissioning stage ~~() occurred~~ in early 2017 runs (5.4 fb^{-1}), where all primary triggers were duplicated ~~with~~ and either the Neural-Ringer or cut-based ~~algorithms operating~~ algorithm operated in the FastCalo stage of the HLT~~;~~.

iii The ~~Operation~~ operation stage, starting after ~~2017 Technical Stop~~Technical Stop 1 (TS1) ~~. Triggers with~~ in 2017. Triggers employing the Neural-Ringer ~~are used as~~ were used as the main triggers, and ~~cut based are~~ cut-based triggers were kept as backup triggers. Final validation of the Neural-Ringer ~~is performed comparing data on~~ was performed by comparing data from this period (39.0 fb^{-1}) as shown in Section 3.1 and Section 4.

iv The Neural-Ringer-only stage, during 2018. Duplicated cut-based triggers ~~are~~ were removed. The performance of the Neural-Ringer in 2018 is presented in Section 3.2.

This section reports the Neural-Ringer's operation in each ~~one~~ of these periods. ~~Additionally, a discussion about CPU time will be covered~~ In addition, CPU-time usage is discussed in Section 3.3.

3.1 2017 ~~Operation~~operation

A backup ~~single electron~~ single-electron HLT_e28_lhtight_nod0_(noringer)_ivarloose trigger, both with and without (noringer) the Neural-Ringer, was employed for monitoring purposes after the first technical stop (TS1). This trigger ~~require~~ requires an electron candidate with ~~$E_T > 28$~~ $E_\mathrm{T} > 28$ GeV satisfying the ~~tight~~ 'tight' selection (lhtight_nod0) with ~~a loose~~ 'loose' isolation criteria (ivarloose) [1]. During the monitoring process, ~~a similar operation of both triggers was observed~~ the two trigger variants were observed to operate similarly in terms of signal efficiency ~~using~~ for the integrated luminosity ~~along~~ during the period, as shown in Figure 6. The trigger turn-on curves exhibit similar ~~profile. In η, the~~ $E_\mathrm{T}$profiles. The Neural-Ringer shows a reasonably symmetric profile ~~with respect to~~ for positive and negative η values. In addition, ~~a~~ an efficiency difference of about ~~half to one percentage point may~~ 0.5% to 1% can be observed in ~~η because of~~ the transition region ~~between the end-cap~~ near $|\eta| = 1.5$ between the endcaps and barrel. ~~Besides aforementioned points, overall efficiency fluctuations are~~ Apart from these features, the overall efficiency differences are typically smaller than a few~~per-mille~~ per mille. Although a slightly more prominent efficiency loss ~~with respect to is observed~~ is observed at large $\langle\mu\rangle$ in Figure 6(c) for the Neural-Ringer trigger, the electron efficiency was kept nearly the same. ~~Such~~ This loss occurs after the linear threshold correction limit of $\langle\mu\rangle = 40$ ~~that~~ was employed during 2017.

Other important triggers were assessed by comparing the trigger efficiencies ~~on~~ in 2017 collision data collected before and after switching to the Neural-Ringer algorithms~~can be observed~~, as shown in Figure 7. The ~~single electron~~ single-electron HLT_e17_lhvloose_nod0_L1EM15VHI trigger ~~require~~ requires an electron candidate with $E_\mathrm{T} > 17$ GeV, seeded by the ~~Level-1~~ first-level trigger L1_EM15VHI~~with $E_T > 17$ GeV~~, and satisfying the loosest identification criteria (lhvloose). On the other hand, HLT_e26_lhtight_nod0_ivarloose ~~and HLT_e60_lhmedium_nod0~~ requires an electron candidate with ~~$E_T > 26$ Gev satisfying the tight~~

$E_T > 26$ GeV satisfying ‘tight’ identification (lhtight) with ~~loose~~ ‘loose’ isolation (ivarloose), and ~~$E_T > 60$ GeV satisfying the medium~~ HLT_e60_lhmedium_nod0 requires an electron candidate with $E_T > 60$ GeV satisfying ‘medium’ identification (lhmedium)~~, respectively. Here, the~~. The electron efficiency was also kept nearly unchanged for ~~relevant single electron triggers. Note that results in this plot are computed with two different data taking periods: with or without~~these relevant single-electron triggers. As mentioned above, the results in Figure 7 are computed for two different data-taking periods: one with, and the other without, the Neural-Ringer algorithm.

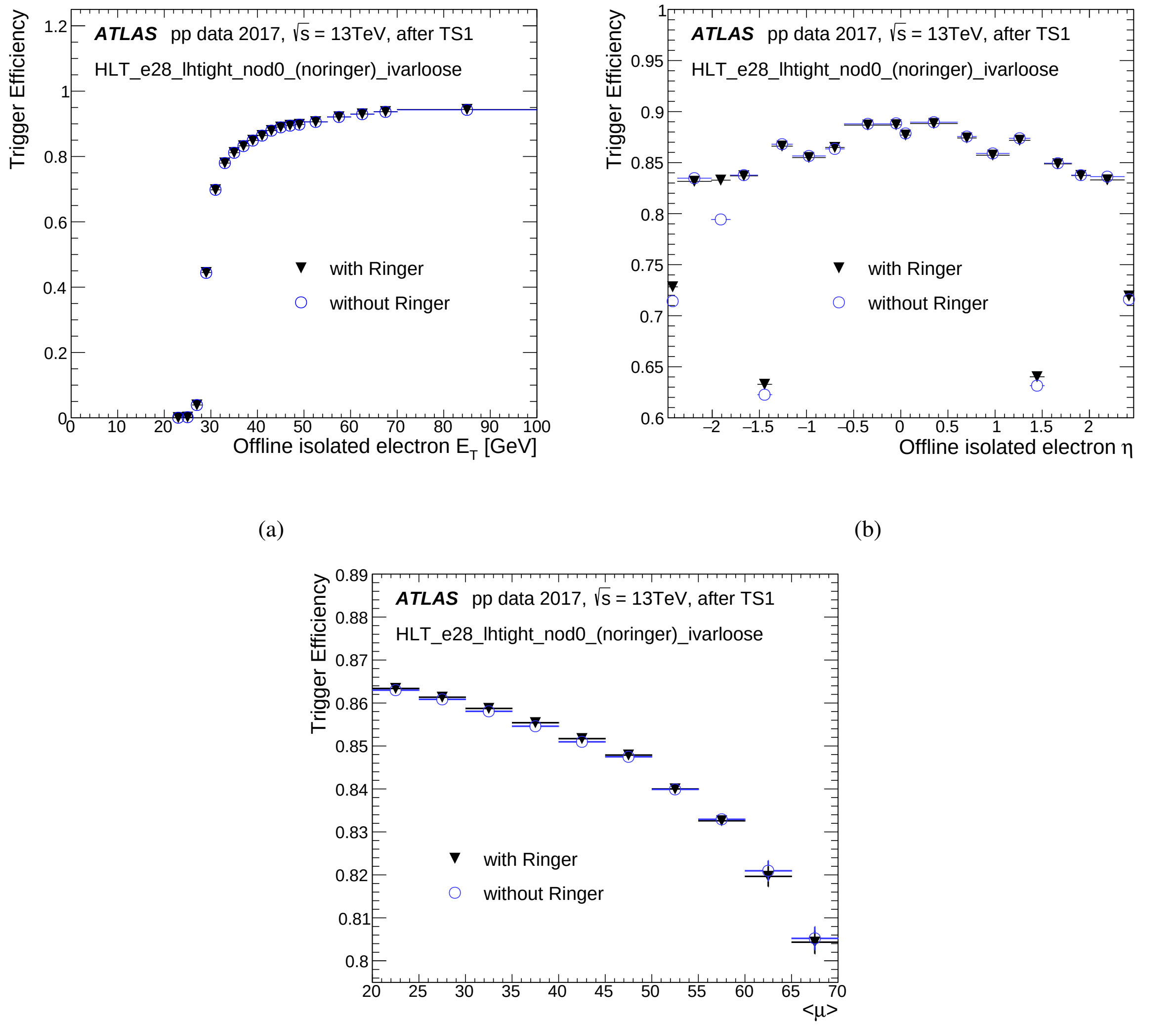
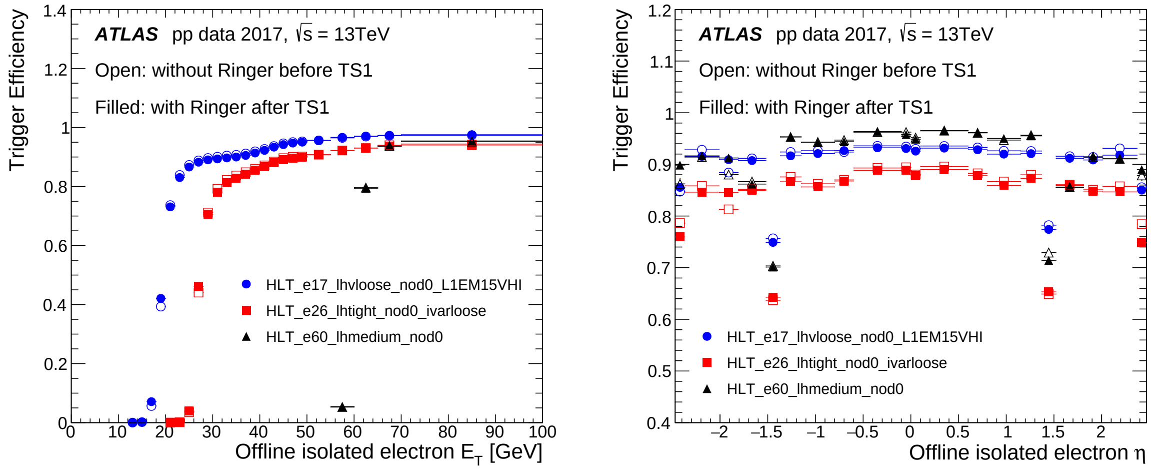


Figure 6: HLT electron efficiency in 2017 collision data as a function of (a) E_T, ~~η~~ (~~e~~b) η, and (~~e~~c) $\langle\mu\rangle$ for the ~~single electron isolated~~ single-isolated-electron trigger requiring $E_T > 28$ GeV and ~~tight~~ ‘tight’ selection with and without the Neural-Ringer algorithm ~~employing 2017 collision data~~ after ~~the~~ its deployment.

~~By contrasting the behavior~~ The power of the Neural-Ringer algorithm is made clear by examining the behaviour of the duplicated trigger ~~using fake electron collision data, it becomes clear the power of the algorithm. An overall reduction factor of the fake rate~~ for fake electron candidates in collision data. The



(a) (b)

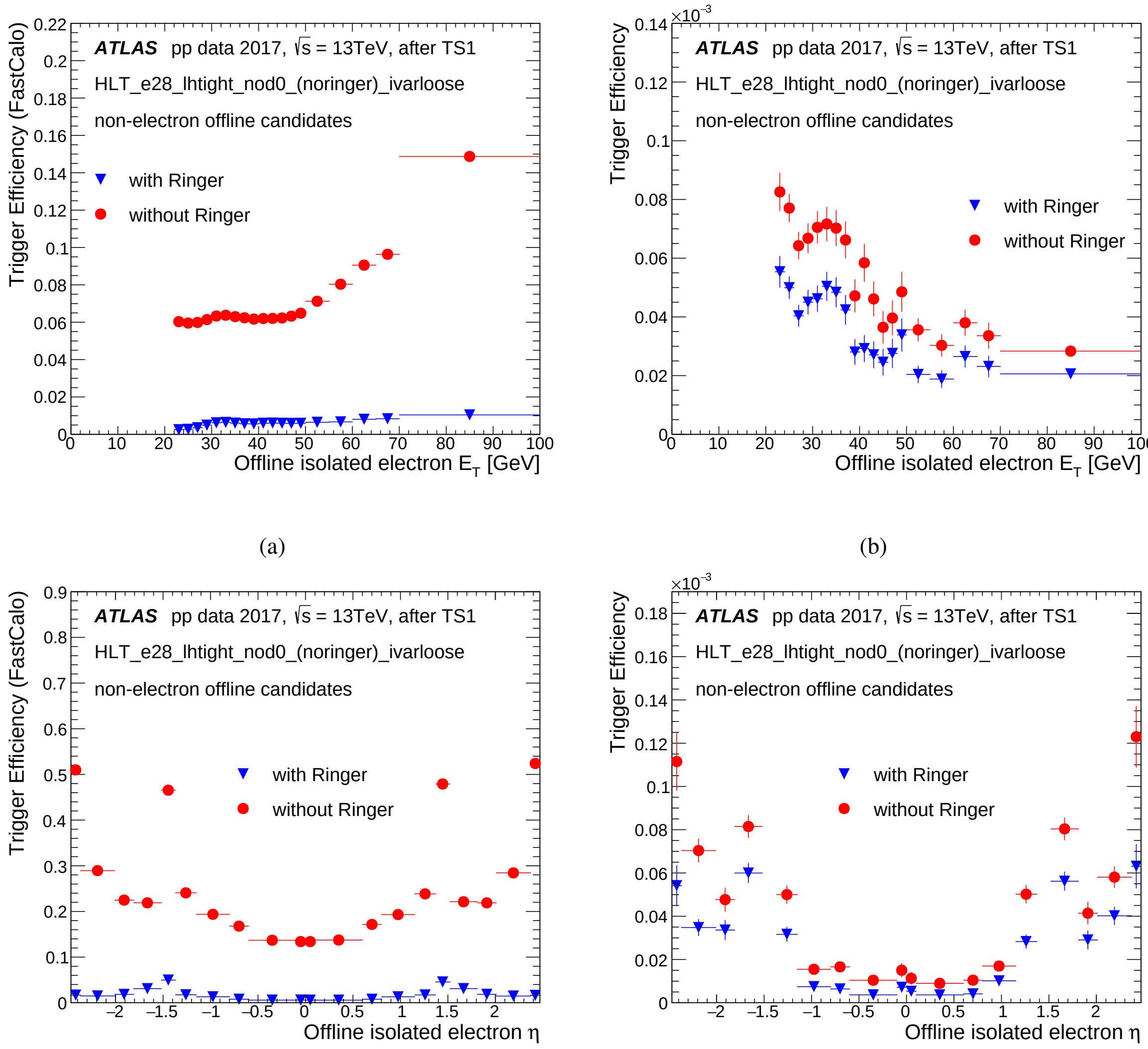
Figure 7: Efficiency of three ~~single electron~~ single-electron triggers as a function of (a) E_{T} and ~~η~~ (b) η for 2017 collision data. Open (filled) markers ~~contain~~ show the efficiency measurements ~~on~~ in runs before (after) the deployment of the Neural-Ringer, thus referring to triggers ~~being~~ executed without (with) the Neural-Ringer algorithm.

overall fake-electron rate is reduced by a factor of 13.75 ~~is achieved at~~ at the FastCalo step. It can be seen ~~in~~ on the left side of Figure 3.1 ~~left side,~~ that the improvement at the FastCalo step is similar for all regions in the evaluated variables, which is particularly interesting when considering the low E_{T} and ~~the end-cap~~ endcap regions. Besides ~~the capability of improving early fake rejection, the usage~~ improving early fake-electron rejection, use of the Neural-Ringer also ~~contributed to reduce the final fake rate at HLT step (~~resulted in a factor of two reduction in the final fake-electron rate from the HLT (see right side of Figure 3.1~~right size)by a factor of 2~~), mostly coming from the ~~transition regions in the calorimeter~~calorimeter's barrel–endcap transition regions.

3.2 2018 ~~Operation~~operation

~~In~~ For 2018 operation, the Neural-Ringer ~~operated with a new tune based on~~ was retuned using collision data. ~~For the lowest-transverse energy-threshold unprescaled trigger , an efficiency improvement of at least one percentage point in central value is observed when comparing both periods, resulting from a~~ The efficiency of the lowest-transverse-energy-threshold unprescaled trigger increased by at least 1%, due to better operation in all selection steps, ~~but, in particular, this is due to improvements from~~ and particularly from improvements in the likelihood tunes~~in the period~~.

~~Despite~~ As well as maintaining high electron efficiency, some unprescaled triggers were observed to achieve a small reduction in ~~fake acceptance was observed~~ fake-electron acceptance at the FastCalo step ~~for some unprescaled triggers~~ in comparison with 2017 operation after TS1. For the ~~lowest-transverse energy-threshold the fake~~ lowest-E_{T}-threshold trigger, the fake-electron acceptance was reduced by a factor ~~by~~ of 1.19~~. On the other hand, for the highest-transverse energy~~ , while for the highest-E_{T}-threshold trigger, a reduction factor ~~by~~ of 1.69 was achieved.



(a) (b)

(c) (d)

Figure 8: Efficiency of selecting background at the FastCalo (~~Left~~left) and ~~at~~ HLT (~~Right~~right) steps as a function of ~~E_T~~ E_{T} (~~Top~~top) and η (~~Bottom~~bottom) ~~for~~ with a ~~single electron isolated~~ single-isolated-electron trigger requiring $E_{\mathrm{T}} > 28$ GeV and ~~tight~~ 'tight' selection, with and without the Neural-Ringer algorithm, employing 2017 collision data collected after the Neural-Ringer algorithm deployment.

3.3 Impact on CPU ~~Demands~~demands

As observed in the efficiency measurements during ~~2017-2018 data taking~~2017–2018 data-taking, the Neural-Ringer allowed ~~for a~~ more effective FastCalo operation ~~in triggers with electrons above with respect to~~ than with the previous cut-based approach for triggers requiring electron E_{T} above 15 GeV.

The overhead required to compute the NN decision was evaluated by running a ~~single electron~~ single-electron trigger with and without the Neural-Ringer algorithm, in two individual non-concurrent executions using the same dedicated node. ~~, was evaluated.~~ It should be mentioned that, in order to reduce implementation ~~efforts~~effort, electron triggers ~~executing the~~ using Neural-Ringer information compute their ~~information~~

decision after the cut-based reconstruction, ~~whose~~ for which a trigger decision is not computed. As a result of this, the Neural-Ringer electron triggers always demand additional FastCalo processing time.

Figures 9 and 10 show ~~the comparison~~ comparisons of the total processing time per call for the feature extraction and ~~hypothesis~~ hypothesis-testing algorithms in the FastCalo ~~respectively (See~~ step respectively (see the related FastCalo ~~gray~~ blocks in Figure 1). The processing time per call does not include the online data preparation time. The measurements ~~were evaluated from~~ are based on a very loose ~~selection electron trigger~~electron trigger, with individual executions in the same dedicated computer node.

The measurement employed events from the enhanced bias (EB) stream [21] typically extracted from ~~one hour acquisition~~ a one-hour acquisition period with a specific trigger menu based only on ~~first level selection and aiming at~~ first-level selections and a goal of collecting about one million background events more likely to be accepted by the HLT. This ~~behavior is obtained by applying an~~ is achieved by applying increased weighting in the high-p_{T} region, and at ~~a~~ an output trigger rate of 300 Hz [22] for one hour. Hence, the measurements are performed under pile-up conditions~~with the execution of the reconstructed observable algorithm~~, with the reconstruction algorithm executing for multiple RoIs in the same bunch-crossing event.

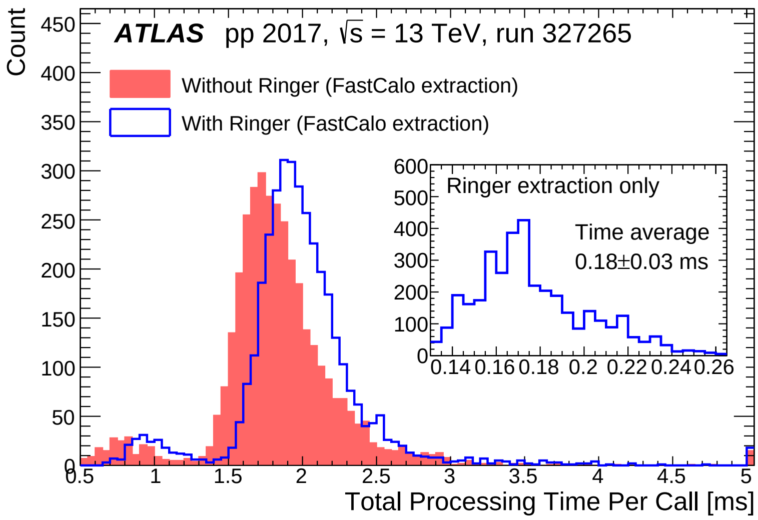


Figure 9: Total processing time per call for the feature extraction algorithms in the FastCalo step of electron ~~offline rerunning~~ triggers rerunning offline, with (blue) and without (red) the Neural-Ringer~~using~~, on EB events (~~$\langle\mu\rangle = 45$~~ peak $\langle\mu\rangle = 45$). ~~Detail~~ The inset on the right shows the ~~individual~~ processing time per call ~~of~~ for only the ring variable extraction.

A multi-modal structure can be observed in the feature extraction distributions of both trigger configurations~~, which is associated to~~; it is associated with the algorithm for retrieving the EM2 ~~cells~~ cell energies and building related ~~shower shape variables[3].~~shower-shape variables.[3] The computation of the ring variables, which is the only difference between the triggers in the FastCalo feature extraction, requires an additional average processing time of 0.18 ± 0.03 ms/event.

[3] ~~The three peak structure comes from the raw data conversion. Particularly, these are amongst the most-demanding contributions to the total CPU time.~~

[3] The three-peak structure comes from the raw data conversion. This makes one of the largest demands contributing to FastCalototal CPU time.

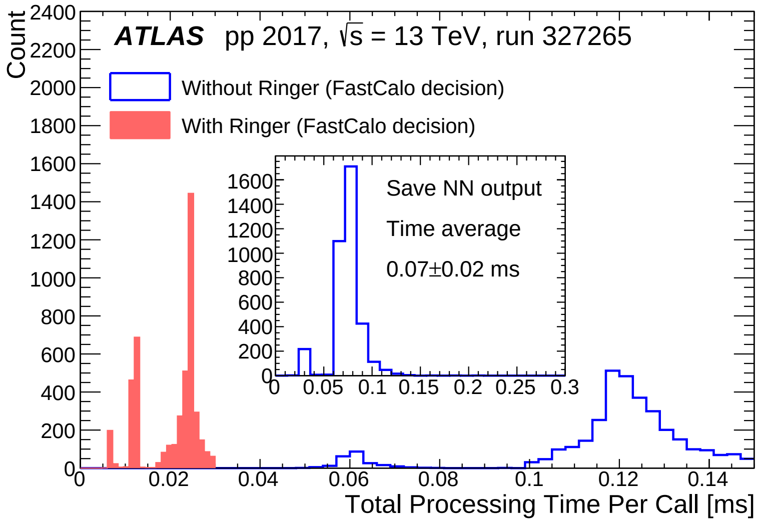


Figure 10: Total processing time per call for the ~~hypothesis testing~~ hypothesis-testing algorithms in the FastCalo step of electron ~~offline rerunning~~ triggers rerunning offline, with (blue) and without (red) the Neural-Ringer~~using~~ , on EB events (~~$\langle\mu\rangle = 45$~~ peak $\langle\mu\rangle = 45$). The ~~center~~ centre histogram represents the ~~individual~~ processing time per call to store the whole Neural-Ringer output in persistent file format.

A ~~less relevant~~ smaller contribution comes from hypothesis testing, in which the ring variables are normalized, the discriminating variable's value is computed, compared ~~to~~ with the selection requirement~~and stored into the~~ , and stored in persistent file format[4] ~~.Specifically, hypothesis testing~~ .[4] Specifically, hypothesis testing for the Neural-Ringer is considerably more ~~demanding,~~ time-demanding (up to 0.14 ms/event, mostly due to ~~the time~~ taking 0.07 ± 0.02 ms/event to store the ~~neural network output,~~ neural-network output) than the cut-based selection (0.02 ms/event), ~~however small with respect to the feature extractionvalues~~but less time-demanding than feature extraction. Additionally, the time increase ~~when compared to~~ is negligible in comparison with the average processing time of a ~~single electron~~ single-electron trigger across the entire HLT ~~is still negligible~~ ($\approx$ 500~~ms in Run 2~~ ms in Run 2).

Taking ~~into account the referred values~~these values into account, the Neural-Ringer can require ~~a relative increase of~~ up to 50% ~~in the~~ more FastCalo CPU time per event ~~with respect to~~ than the trigger without the Neural-Ringer[5] ~~.~~.[5] Nonetheless, the Neural-Ringer ~~contributes to~~ provides a more discriminating selection, ~~allowing to reduce CPU demanding~~ which reduces the number of CPU-demanding triggers by reducing the ~~processing of~~ need to process fake electrons in subsequent steps that are more computationally demanding. In this measurement, where about ~~3,400~~ 3400 events from the EB stream were considered, the Neural-Ringer algorithm provided an overall 60% reduction ~~(~~in the CPU demands (from 30.72 ms to 10.36 ms ~~) in the CPU demands avoiding the computation of fake candidates after~~ per event) by rejecting most fake electron candidates during the FastCalo step.

[4] ~~For Run-3 this step has been removed as a way to reduce the CPU time at step.~~

[4] For Run 3, this step has been removed as a way to reduce the CPU time at the FastCalostep.

[5] ~~It is expected that the result be dependent on the trigger configuration due to presence of pile-up. Reported results are for HLT_e17_lhvloose_nod0 single electron trigger.~~

[5] The CPU-time increase is expected to depend on the trigger configuration because of the presence of pile-up. The reported results are for the HLT_e17_lhvloose_nod0 single-electron trigger.

4 Neural-Ringer ~~Aggreament~~assessment vs cut-based triggers

This section ~~is dedicated to~~ presents a more detailed ~~assessments~~ assessment of the Neural-Ringer~~behavior~~ 's behaviour in the period ~~2017-2018 with respect~~ 2017–2018 relative to the previous electron trigger's cut-based strategy. ~~Using~~ A $Z \to ee$ tag-and-probe selection ~~, the~~ was used to evaluate the level of agreement between the two triggers~~and possible biases , caused by the introduction of~~, as well as possible biases due to the variables the Neural-Ringer ~~using the variables employed by the likelihood algorithm, were investigated. Since~~ employs in the likelihood algorithm. Since both the offline and final HLT electron selections are based on such variables, limiting the evaluation to ~~them suffices to understand any~~ those varaibles suffices for understanding the possible impact of the Neural-Ringer algorithm in most analyses. ~~Additionally, these variables are interesting since they~~ In addition, it is advantageous to compact the input space ~~in~~ into a set of ~~few variables with~~ a few such variables that have low correlation and ~~high interpretability power~~are easy to interpret.

Since ~~the Tag & Probe events have~~ each tag-and-probe event has one electron, called ~~as probe electron~~the probe electron, that was not used to ~~decide the selection of~~ select that event at ~~the~~ trigger level, two offline analyses were performed to assess the trigger performance with and without the Neural-Ringer, ~~when the selection is applied~~ with the offline selection applied either to the tag electron (Homogeneity Tests) or to the probe electron (Quadrant Analysis).

The Quadrant Analysis allows ~~to directly assess possible bias caused by~~ a direct assessment of possible bias due to the introduction of the Neural-Ringer by comparing the profiles for all possible disjoint decisions from the two trigger classifiers. ~~Hence, it~~ It is possible for the classifiers to agree~~by both~~, with both of them accepting or rejecting the event. Likewise, they ~~are able to~~ can disagree in two possible ~~cases where either one decides to accept the event while the other rejects~~ ways, with one classifier accepting the event and the other rejecting it.

Although it is reasonable to ~~expect that the shower development of the tag and the probe are independent from each other~~assume that the tag and probe showers develop independently, and that the Neural-Ringer when applied to the tag selection cannot, in principle, alter the profile of the probes besides changing the number of tag-and-probe pairs~~. Such~~, this assumption should be tested. An evaluation of any possible systematic effect ~~caused by the introduction of the~~ due to using the Neural-Ringer in ~~the extraction of the likelihood PDFs, employed~~ obtaining the likelihood probability distribution functions (PDFs), used for offline and final HLT selection, is performed ~~by~~ with the Homogeneity Tests.

4.1 Quadrant Analysis

The Quadrant Analysis was developed to compare the ~~decision~~ decisions of two classifiers. Given signal candidates and two classifiers, these are the possibilities: either both accept or reject the candidates, or only one accepts the candidates. The four possible decision quadrants are evaluated from ~~shower shape~~ shower-shape distributions. Furthermore, in this analysis, it ~~'s~~ is also possible to ~~compare~~ evaluate the disagreement between two classifiers. The disagreement between two classifiers **A** and **B** can be defined as:

$$\text{disagreement} = \frac{N_{\mathbf{A}} + N_{\mathbf{B}}}{N_{\mathbf{A}} + N_{\mathbf{B}} + N_{\mathbf{AB}} + N_{!\mathbf{AB}}} \tag{2}$$

where $N_{\mathbf{A}}$ ($N_{\mathbf{B}}$) is the total number of candidates accepted exclusively by classifier **A** (**B**) and $N_{\mathbf{AB}}$ ($N_{\mathbf{!AB}}$) is the total number of candidates accepted (rejected) by both classifiers.

In this approach, only collision data collected ~~by~~ using the primary triggers with an equivalent offline selection are considered[6]~~.~~.[6] Results in Section 4.1.1 show trigger selection performance using the 'tight' requirement for the duplicated triggers during 2017.

4.1.1 Results

The Quadrant Analysis was performed on the ~~variables (R_η, R_ϕ, E_{ratio} and R_{had}~~four variables (R_η, R_ϕ, E_{ratio} and R_{had}) used in the likelihood [2] selection ~~and~~ that have the best discrimination power between electrons and jets.

As shown in Figure ~~11(a)~~11, the disagreement is small and ~~bounded for most cases at~~ in most cases bounded at about the 1% level for the coverage of ~~all calorimetry variables. This behavior~~ each variable. Here, only the results for $0.6 < |\eta| < 0.8$ and $35 < E_T < 40$ GeV are shown, but this behaviour is also observed for other E_T regions, and these results are more representative of those obtained in other regions of the $E_T \times |\eta|$ phase space. One should note that the working points of ~~both~~ the two triggers are not exactly the same, although the Neural-Ringer aims ~~at keeping~~ to achieve the same signal efficiency as the cut-based strategy and was operating as desired. In other words, the ~~difference in height of~~ height differences between the blue and red profiles in Figure ~~11(a) is~~ 11 are mainly due to ~~the small differences in efficiencies~~ small efficiency differences between the two triggers. ~~Here, only the results for $0.6 < |\eta| < 0.8$ and $35 < E_T[\text{GeV}] < 40$ are shown, since these results are more representative of the other results obtained in the rest of the regions of the $E_T \times |\eta|$ phase space.~~

~~For all variables in this region, both~~ The two triggers behave very similarly ~~(see~~ for all variables in the region shown in Figure 11~~)~~, even if some slight shifts ~~can be observed in few profiles~~between blue and red profiles can be seen. This is the case ~~of~~ for R_η and R_ϕ, where the Neural-Ringer trigger ~~is consistently collecting~~ consistently selects slightly more events in the signal region, i.e.~~respectively with slight~~ ones with slightly tighter showers in η and ϕ, respectively, in the EM2 layer. The Neural-Ringer trigger~~behavior~~'s behaviour shows an even ~~lower effect in the~~ smaller effect in R_{had}, where tighter tails are observed, resulting in fewer electrons with 10% to 30% of their energy in EM1, 1% to 2% in EM3 and more than 4 GeV hadronic leakage. The E_{ratio} ~~also shows a slightly bias towards collecting~~ variable also shows the Neural-Ringer trigger has a slight bias towards selecting more events in the signal region~~with the trigger~~. Interestingly, a few events with $0.5 < E_{\text{ratio}} < 0.7$ are accepted by the trigger with the Neural-Ringer, ~~when $0.5 < E_{\text{ratio}} < 0.7$,~~ probably due to electrons resulting from premature showers with ~~barycenter near the edge of~~ a barycentre near the boundary between two strips. Although similar ~~behavior is shown for the~~ behaviour is seen in other regions, the differences vary in strength ~~in each~~ between $|\eta|$ ~~region~~regions.

[6] ~~I.e., if a trigger leg is being evaluated, it also applied the offline selection to the candidate.~~

[6] For example, if a 'tight' trigger leg is being evaluated, the 'tight' offline selection is also applied to the candidate.

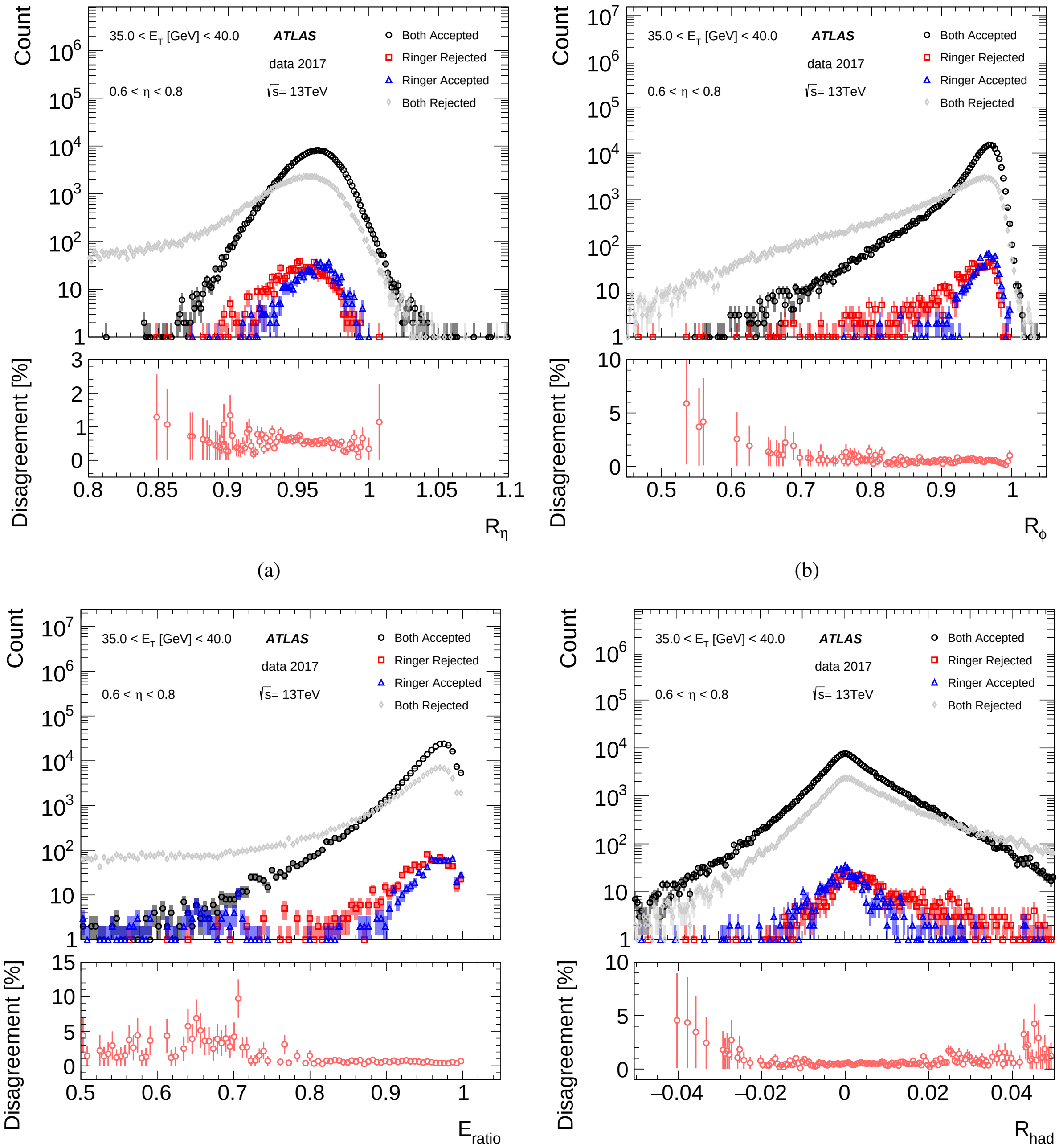


Figure 11: Quadrant ~~analysis~~ Analysis plots for the main offline-reconstructed calorimetry variables employed in the likelihood and for the $0.6 < |\eta| < 0.8$ and ~~$30 < E_{\mathrm{T}}$ [GeV] < 35~~ $30 < E_{\mathrm{T}} < 35$ GeV region. The top pad in each figure shows the raw number of ~~observations~~ candidates for the four mutually exclusive cases: accepted both with and without the Neural-Ringer ~~triggered~~ (black); ~~triggered~~ accepted only with the Neural-Ringer (blue); ~~triggered~~ accepted only without the Neural-Ringer (red); ~~neither one triggered~~ rejected both with and without the Neural-Ringer (~~gray~~grey). The bottom pad ~~contains~~ shows the disagreement as defined in ~~the Equation~~ Eq. (2).

4.2 Homogeneity Tests

In this analysis, the impact on the likelihood algorithm of using different triggers, Neural-Ringer and cut-based, ~~on the likelihood algorithm~~ is evaluated. This study requires a pair of duplicated triggers, ~~composed by an $E_T > 28$~~ applying an $E_{\mathrm{T}} > 28$ GeV isolated and ~~Tight~~ 'tight' electron selection with and without the Neural-Ringer, operating ~~online together . This~~ together online. The analysis is performed using $Z \to ee$ tag-and-probe collision data, which were selected with the trigger requirement set to either one of the duplicated online triggers for the tag, whereas the probe selection is invariably set to the offline VeryLoose criterion. This is ~~the exact setup employed by ATLAS~~ exactly the same set-up that ATLAS employed to derive the offline likelihood for electrons with E_{T} above 15 GeV [2]. To benefit from the full 2017 ~~statistics~~dataset, collision data collected ~~previous to the~~ before TS1 were also employed. ~~In case~~ If the profiles are statistically identical, ~~then it is expected that~~ the Neural-Ringer trigger ~~does not~~ is not expected to cause any relevant ~~alteration~~ change in the derivation of the offline likelihood ~~pdfs~~PDFs.

The problem is ~~approached based on homogeneity tests on~~ addressed by applying homogeneity tests to histograms [23] ~~,~~ in order to check for systematic effects of the trigger configuration. ~~It is~~ These tests are based on a test originally developed by Pearson [24] and popularly employed in many fields beyond ~~High Energy Physics (HEP)~~high-energy physics, e.g. social sciences [25] and health [26], ~~usually labelled as~~ where they are usually called contingency or consistency tests.

In order to benefit from the ~~test~~ tests without having to customize ~~it to the ATLAS particular analysis setup~~them to an ATLAS-specific analysis set-up, the collision data ~~was~~ were split into two statistically independent groups~~attempting to keep data taking~~ , consisting of data alternately taken from consecutive short time periods to keep their data-taking conditions as similar as possible~~by successively taking data to each group from small consecutive periods~~.

The ~~p-value~~p-value, or probability value, is a measure used in statistical hypothesis testing to determine the significance of ~~your~~ one's results. It represents the probability of obtaining an observed result, or something more extreme, if the null hypothesis (which states that there is no effect or no difference) is true. By comparing the ~~p-values~~ p-values of the two groups, it is possible to evaluate how the different configurations affect the likelihood ~~of the profiles~~ that the profiles are being drawn from the same distribution. If the ~~p-values~~ p-values have negligible fluctuations with respect to the values obtained ~~by the~~ from tests comparing the same triggers, then the systematic effect in the profiles is negligible ~~with respect to half~~for a dataset that is half[7] of the ~~statistics available~~ size of the whole available dataset.

To provide ~~a better insight on~~ better insight into possible distortions, the corresponding ~~χ individual~~ individual χ contributions of each group are computed, allowing them to ~~freely oscillate through the positive and negative axis with~~ assume positive or negative values by defining

$$\chi^{s}_{i,j} = \frac{(r_{i,j} - b_{i,j})}{\sqrt{b_{i,j}}}, \tag{3}$$

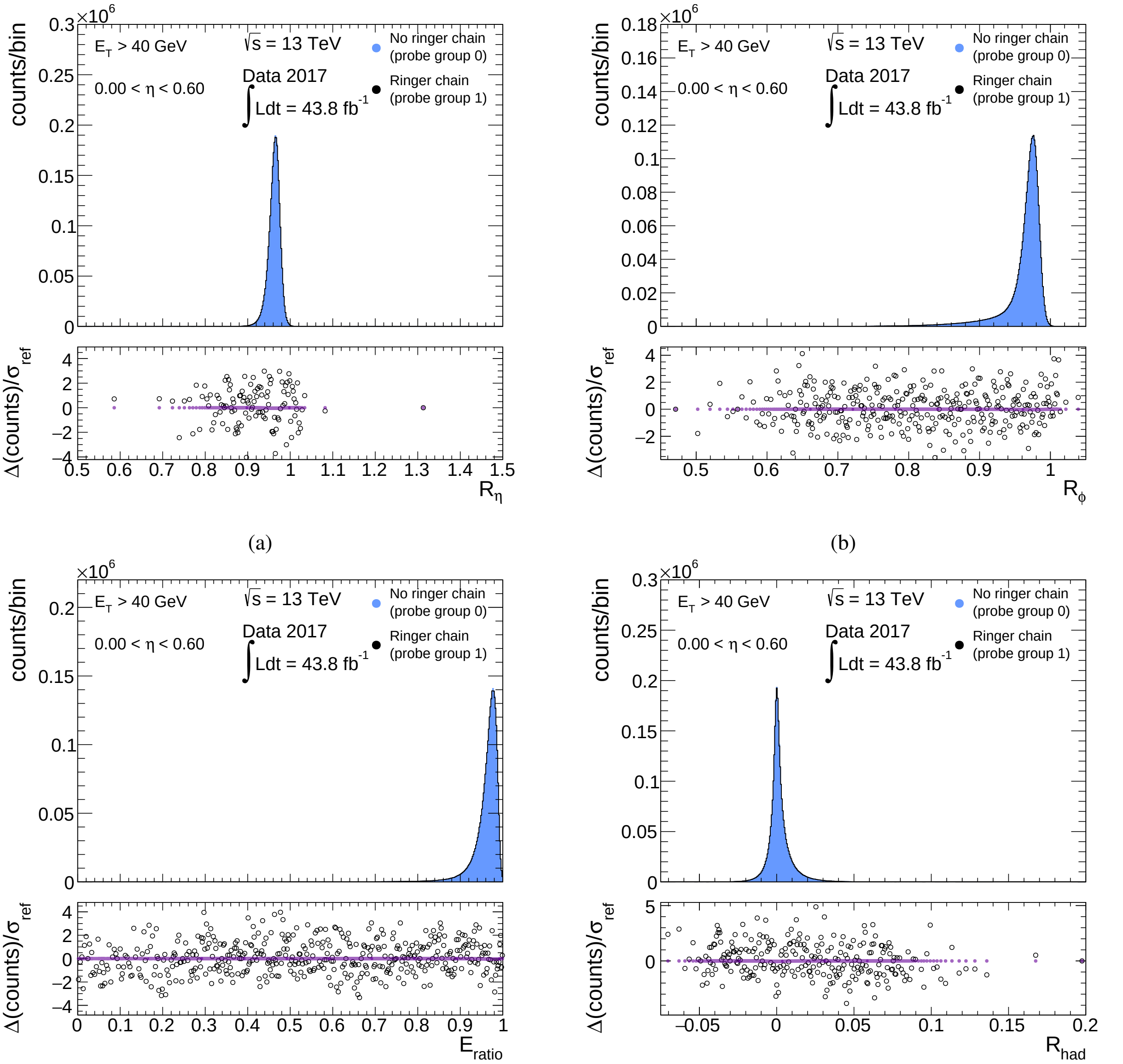
where $r_{i,j}$ ($b_{i,j}$) is the number of observations collected by the trigger with (without) the Neural-Ringer in ~~jth~~ j^{th} histogram bin and in the ~~ith~~ i^{th} $E_{\mathrm{T}} \times |\eta|$ region.

7 ~~Once each~~ Each group ~~is using nearly~~ contains approximately half of the integrated luminosity.

4.2.1 Results

Regardless the trigger configuration, the ~~p-values of the~~ homogeneity tests between the two groups result in similar ~~values~~p-values. In other words, the fluctuations in the profiles are mainly ~~dominated by statisticalfluctuations~~statistical, with no sign of systematic effects due to the trigger configuration.

~~A similar behavior is observed for the ID and calo-ID variables. The~~ Figure 12 shows the residuals ~~when considering the histogram obtained with~~ for the R_{η}, R_{ϕ}, E_{ratio}, and R_{had}calorimetry variables when the reference histogram is obtained from collision data collected by the trigger without the Neural-Ringer for the first ~~arbitrary groupas a reference~~group. The residuals are dominated by statistical fluctuations, ~~which~~ and are within the expectations from the homogeneity hypothesis for most variables and regions. Similar behaviour is observed for track variables.



(a) (b) (c) (d)

Figure 12: Top: Profile histograms for the R_η, ~~,~~ R_ϕ, E_{ratio}, and R_{had} ~~histogram profiles for the~~ calorimetry variables employed in the offline likelihood in the $E_T > 40$ GeV and ~~$0.00 < |\eta| < 0.80$~~ $0.0 < |\eta| < 0.8$ regions using collision data collected by triggering without the Neural-Ringer in the first arbitrary data group (blue area histogram) and collision data collected by triggering with the Neural-Ringer in the second arbitrary data group (black line histogram). Bottom: residual contributions using ~~as statistics~~ the χ^s statistic (~~equation~~black circles, from Eq. (3~~, in black~~)), and the expected ~~model for no distortion given by the~~ χ^s residuals ~~w.r.t.~~ if there are no distortions or statistical fluctuations with respect to the reference (purple dots at zero).

5 Conclusion

The Neural-Ringer ~~is~~ has an innovative design for electron selection based on calorimetry information. With the ~~ambition to comply with~~ goal of satisfying specific trigger system demands, ~~the feature extraction~~

~~of~~ feature extraction from concentric ring energy sums provides an efficient description, which is quickly derived through simple operations. ~~Neural network~~ Neural-network models are employed ~~for exploiting nonlinear~~ to exploit non-linear correlations between rings~~and the optimisation~~, and the optimization of this strategy culminated in the ~~employment~~ deployment of a set of ~~specialist models per regions~~ models specific to each region of pseudorapidity and transverse energy.

In the second half of 2017, the Neural-Ringer algorithm replaced ~~such~~ a cut-based strategy in the FastCalo decision step of the HLT and became the baseline algorithm for triggering events containing electrons with E_T above 15 GeV, as part of the ATLAS online system~~CPU~~'s CPU-time reduction campaign. ~~furthermore~~ Furthermore, it was the first time a ~~neural network~~ neural-network method was employed as a baseline ~~selection algorithm for event selection~~ event-selection algorithm in the ATLAS trigger system.

During 2017 operation, ~~based on neural networks trained with~~ with the Neural-Ringer algorithm trained on simulated data, ~~a reduction factors~~ fake-electron trigger rates were reduced by a factor of 13 in the FastCalo ~~and~~ step and by a factor of 2 overall in the HLT ~~of fake rates were achieved~~ with respect to the offline likelihood ~~, while keeping high electron efficiency similar to lowest energy-threshold unprescaled single electron trigger~~algorithm, while achieving high electron-selection efficiencies very similar to those of relevant single-electron triggers without the Neural-Ringer. For 2018, the Neural-Ringer was trained with collision data, resulting in an estimated additional reduction factor of 1.19–1.69 for fake-electron triggers in the FastCalo step with respect to 2017. ~~Additionally, the~~ The Neural-Ringer was also studied ~~through the~~ from an offline likelihood perspective to ~~bring additional insights on its behavior~~ obtain additional insights into its behaviour by using two equivalent ~~single electron trigger~~ single-electron triggers with and without the Neural-Ringer. The ~~proposed quadrant analysis allowed to observe high agreement between both triggerswith slight shifts~~ 'quadrant analysis' showed a high level of agreement between the two triggers, with the Neural-Ringer trigger showing a slight bias towards the signal region in the disagreement profiles of some calorimetry-based variables. Finally, ~~when considering homogeneity test, the p-values of the~~ the 'homogeneity tests' between the two groups ~~result in similar values~~of data resulted in similar p-values. In other words, the fluctuations in the profiles are mainly ~~dominated by statisticalfluctuations~~statistical, with no sign of systematic effects due to the trigger configuration.

Physics beyond the Standard Model is getting more attention at the LHC and is expected to be part of the main focus in the data-taking phases to come. At the beginning of ~~Run 3~~Run 3 , the Neural-Ringer was updated to ~~comprise closer-by electron (~~include events in which electrons have smaller angular separations (such as in boosted topologies)~~events~~, particularly limiting the effect of other contributions at the edge of the feature extraction window, which can result from these physics processes. In addition, ~~an extension to~~ the Neural-Ringer was extended into the $E_\mathrm{T}$region below 15~~GeV was implemented~~ GeV, motivated by the high CPU ~~demand of all~~ demands of triggers in this kinematic region.

Acknowledgements

We thank CERN for the very successful operation of the LHC and its injectors, as well as the support staff at CERN and at our institutions worldwide without whom ATLAS could not be operated efficiently.

The crucial computing support from all WLCG partners is acknowledged gratefully, in particular from CERN, the ATLAS Tier-1 facilities at TRIUMF/SFU (Canada), NDGF (Denmark, Norway, Sweden), CC-IN2P3 (France), KIT/GridKA (Germany), INFN-CNAF (Italy), NL-T1 (Netherlands), PIC (Spain), RAL (UK) and BNL (USA), the Tier-2 facilities worldwide and large non-WLCG resource providers. Major contributors of computing resources are listed in Ref. [27].

We gratefully acknowledge the support of ANPCyT, Argentina; YerPhI, Armenia; ARC, Australia; BMWFW and FWF, Austria; ANAS, Azerbaijan; CNPq and FAPESP, Brazil; NSERC, NRC and CFI, Canada; CERN; ANID, Chile; CAS, MOST and NSFC, China; Minciencias, Colombia; MEYS CR, Czech Republic; DNRF and DNSRC, Denmark; IN2P3-CNRS and CEA-DRF/IRFU, France; SRNSFG, Georgia; BMBF, HGF and MPG, Germany; GSRI, Greece; RGC and Hong Kong SAR, China; ISF and Benoziyo Center, Israel; INFN, Italy; MEXT and JSPS, Japan; CNRST, Morocco; NWO, Netherlands; RCN, Norway; MNiSW, Poland; FCT, Portugal; MNE/IFA, Romania; MESTD, Serbia; MSSR, Slovakia; ARRS and MIZŠ, Slovenia; DSI/NRF, South Africa; MICINN, Spain; SRC and Wallenberg Foundation, Sweden; SERI, SNSF and Cantons of Bern and Geneva, Switzerland; MOST, Taipei; TENMAK, Türkiye; STFC, United Kingdom; DOE and NSF, United States of America.

Individual groups and members have received support from BCKDF, CANARIE, CRC and DRAC, Canada; CERN-CZ, FORTE and PRIMUS, Czech Republic; COST, ERC, ERDF, Horizon 2020, ICSC-NextGenerationEU and Marie Skłodowska-Curie Actions, European Union; Investissements d'Avenir Labex, Investissements d'Avenir Idex and ANR, France; DFG and AvH Foundation, Germany; Herakleitos, Thales and Aristeia programmes co-financed by EU-ESF and the Greek NSRF, Greece; BSF-NSF and MINERVA, Israel; NCN and NAWA, Poland; La Caixa Banking Foundation, CERCA Programme Generalitat de Catalunya and PROMETEO and GenT Programmes Generalitat Valenciana, Spain; Göran Gustafssons Stiftelse, Sweden; The Royal Society and Leverhulme Trust, United Kingdom.

In addition, individual members wish to acknowledge support from Armenia: Yerevan Physics Institute (FAPERJ); CERN: European Organization for Nuclear Research (CERN PJAS); Chile: Agencia Nacional de Investigación y Desarrollo (FONDECYT 1230812, FONDECYT 1230987, FONDECYT 1240864); China: Chinese Ministry of Science and Technology (MOST-2023YFA1605700), National Natural Science Foundation of China (NSFC - 12175119, NSFC 12275265, NSFC-12075060); Czech Republic: Czech Science Foundation (GACR - 24-11373S), Ministry of Education Youth and Sports (FORTE CZ.02.01.01/00/22_008/0004632), PRIMUS Research Programme (PRIMUS/21/SCI/017); EU: H2020 European Research Council (ERC - 101002463); European Union: European Research Council (ERC - 948254, ERC 101089007), Horizon 2020 Framework Programme (MUCCA - CHIST-ERA-19-XAI-00), European Union, Future Artificial Intelligence Research (FAIR-NextGenerationEU PE00000013), Italian Center for High Performance Computing, Big Data and Quantum Computing (ICSC, NextGenerationEU); France: Agence Nationale de la Recherche (ANR-20-CE31-0013, ANR-21-CE31-0013, ANR-21-CE31-0022), Investissements d'Avenir Labex (ANR-11-LABX-0012); Germany: Baden-Württemberg Stiftung (BW Stiftung-Postdoc Eliteprogramme), Deutsche Forschungsgemeinschaft (DFG - 469666862, DFG - CR 312/5-2); Italy: Istituto Nazionale di Fisica Nucleare (ICSC, NextGenerationEU); Japan: Japan Society for the Promotion of Science (JSPS KAKENHI JP22H01227, JSPS KAKENHI JP22H04944, JSPS KAKENHI JP22KK0227, JSPS KAKENHI JP23KK0245); Netherlands:

Netherlands Organisation for Scientific Research (NWO Veni 2020 - VI.Veni.202.179); Norway: Research Council of Norway (RCN-314472); Poland: Ministry of Science and Higher Education (IDUB AGH, POB8, D4 no 9722), Polish National Agency for Academic Exchange (PPN/PPO/2020/1/00002/U/00001), Polish National Science Centre (NCN 2021/42/E/ST2/00350, NCN OPUS nr 2022/47/B/ST2/03059, NCN UMO-2019/34/E/ST2/00393, UMO-2020/37/B/ST2/01043, UMO-2021/40/C/ST2/00187, UMO-2022/47/O/ST2/00148, UMO-2023/49/B/ST2/04085, UMO-2023/51/B/ST2/00920); Slovenia: Slovenian Research Agency (ARIS grant J1-3010); Spain: Generalitat Valenciana (Artemisa, FEDER, IDIFEDER/2018/048), Ministry of Science and Innovation (MCIN & NextGenEU PCI2022-135018-2, MICIN & FEDER PID2021-125273NB, RYC2019-028510-I, RYC2020-030254-I, RYC2021-031273-I, RYC2022-038164-I); Sweden: Swedish Research Council (Swedish Research Council 2023-04654, VR 2018-00482, VR 2022-03845, VR 2022-04683, VR 2023-03403, VR grant 2021-03651), Knut and Alice Wallenberg Foundation (KAW 2018.0458, KAW 2019.0447, KAW 2022.0358); Switzerland: Swiss National Science Foundation (SNSF - PCEFP2_194658); United Kingdom: Leverhulme Trust (Leverhulme Trust RPG-2020-004), Royal Society (NIF-R1-231091); United States of America: U.S. Department of Energy (ECA DE-AC02-76SF00515), Neubauer Family Foundation.

References

[1] ATLAS Collaboration, *Performance of electron and photon triggers in ATLAS during LHC Run 2*, Eur. Phys. J. C **80** (2020) 47, arXiv: 1909.00761 [hep-ex] (cit. on pp. 3, 4, 10, 14).

[2] ATLAS Collaboration, *Electron reconstruction and identification in the ATLAS experiment using the 2015 and 2016 LHC proton–proton collision data at $\sqrt{s} = 13$ TeV*, Eur. Phys. J. C **79** (2019) 639, arXiv: 1902.04655 [physics.ins-det] (cit. on pp. 3, 12, 21, 23).

[3] ATLAS Collaboration, *Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC*, Phys. Lett. B **716** (2012) 1, arXiv: 1207.7214 [hep-ex] (cit. on p. 3).

[4] ATLAS Collaboration, *Measurement of the Higgs boson mass in the $H \rightarrow ZZ^* \rightarrow 4\ell$ and $H \rightarrow \gamma\gamma$ channels with $\sqrt{s} = 13$ TeV pp collisions using the ATLAS detector*, Phys. Lett. B **784** (2018) 345, arXiv: 1806.00242 [hep-ex] (cit. on p. 3).

[5] *Technical Design Report for the Phase-II Upgrade of the ATLAS TDAQ System*, tech. rep., CERN, 2017, URL: https://cds.cern.ch/record/2285584 (cit. on p. 3).

[6] ATLAS Collaboration, *The ATLAS Experiment at the CERN Large Hadron Collider*, JINST **3** (2008) S08003 (cit. on p. 5).

[7] ATLAS Collaboration, *Readiness of the ATLAS liquid argon calorimeter for LHC collisions*, Eur. Phys. J. C **70** (2010) 723, arXiv: 0912.2642 [hep-ex] (cit. on p. 5).

[8] ATLAS Collaboration, *ATLAS liquid-argon calorimeter: Technical Design Report*, CERN-LHCC-96-041, 1996, URL: https://cds.cern.ch/record/331061 (cit. on p. 5).

[9] ATLAS Collaboration, *Operation and performance of the ATLAS Tile Calorimeter in Run 1*, Eur. Phys. J. C **78** (2018) 987, arXiv: 1806.02129 [hep-ex] (cit. on p. 5).

[10] ATLAS Collaboration, *ATLAS tile calorimeter: Technical Design Report*, CERN-LHCC-96-042, 1996, URL: https://cds.cern.ch/record/331062 (cit. on p. 5).

[11] ATLAS Collaboration, *ATLAS Calorimeter Performance: Technical Design Report*, CERN-LHCC-96-40, 1996, URL: http://cds.cern.ch/record/331059 (cit. on p. 5).

[12] J. Seixas, L. Caloba, M. Souza, A. Braga and A. Rodrigues, *Neural second-level trigger system based on calorimetry*, Comput. Phys. Commun. **95** (1996) 143, ISSN: 0010-4655 (cit. on pp. 8, 9).

[13] I. Bird et al., *Update of the Computing Models of the WLCG and the LHC Experiments*, tech. rep. CERN-LHCC-2014-014. LCG-TDR-002, 2014, URL: https://cds.cern.ch/record/1695401 (cit. on p. 9).

[14] R. Polikar, *Ensemble based systems in decision making*, IEEE Circuits Syst. Mag. **6** (2006) 21, ISSN: 1531636X, arXiv: arXiv:1011.1669v3, URL: http://ieeexplore.ieee.org/document/1688199/ (cit. on p. 9).

[15] Z.-H. Zhou, *Ensemble Methods: Foundations and Algorithms*, 1st, Chapman & Hall/CRC, 2012, ISBN: 1439830037, 9781439830031 (cit. on p. 9).

[16] S. Haykin, *Neural Networks and Learning Machines*, 3rd ed., Upper Saddle River, NJ, USA: Prentice Hall, 2008 (cit. on pp. 10, 11).

[17] D. Nguyen and B. Widrow, 'Improving the Learning Speed of 2-Layer Neural Networks by Choosing Initial Values of the Adaptive Weights', *Proceeding of the International Joint Conference on Neural Networks*, vol. 3, San Diego, USA, 1990 21 (cit. on p. 11).

[18] M. Riedmiller and H. Braun, 'A Direct Adaptive Method for Faster Backpropagation Learning: The RPROP Algorithm', *Proceedings of the IEEE International Conference on Neural Networks*, San Francisco, USA, 1993 586 (cit. on p. 11).

[19] I. Goodfellow, Y. Bengio and A. Courville, *Deep learning*, 2016 800, URL: http://www.deeplearningbook.org (cit. on p. 11).

[20] D. P. Kingma and J. Ba, *Adam: A method for stochastic optimization*, arXiv preprint arXiv:1412.6980 (2014) (cit. on p. 13).

[21] ATLAS Collaboration, *Description of Enhance-bias stream and the datasets*, URL: https://twiki.cern.ch/twiki/bin/view/Atlas/EnhancedBiasData (cit. on p. 18).

[22] ATLAS Collaboration, *Specifications for Collecting Data with Enhanced Bias Streams*, URL: https://twiki.cern.ch/twiki/bin/view/Atlas/TakingNewEnhancedBiasData (cit. on p. 18).

[23] N. D. Gagunashvili, *Comparison of weighted and unweighted histograms*, ArXiv Physics e-prints (2006), arXiv:physics/0605123 (cit. on p. 23).

[24] K. Pearson, *On the probability that two independent distributions of frequency are really samples from the same population*, Biometrika **8** (1911) 250 (cit. on p. 23).

[25] T. D. Wickens, *Multiway contingency tables analysis for the social sciences*, Psychology Press, 2014 (cit. on p. 23).

[26] C. Ma, G. Shan and S. Liu, *Homogeneity test for correlated binary data*, PloS one **10** (2015) e0124337 (cit. on p. 23).

[27] ATLAS Collaboration, *ATLAS Computing Acknowledgements*, ATL-SOFT-PUB-2023-001, 2023, URL: https://cds.cern.ch/record/2869272 (cit. on p. 27).